KING FAHD UNIVERSITY OF PETROLEUM & MINERALS

Senior Design Project II

Term # 252

Team # F040

Project Title: **Infrared Methane Leak Detection in Pipelines with AI-Enabled Cloud Monitoring**

Final Design Report (FDR)

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Abstract

Methane leakage from pipeline systems poses significant safety, environmental, and economic risks, requiring reliable and timely detection methods. This project presents the design and implementation of a bench-top methane leak detection system based on infrared absorption principles. Infrared light is transmitted across controlled leak regions of a pipeline section, where the presence of methane is detected through attenuation of specific wavelengths caused by gas absorption.

A laboratory pipeline test rig with controlled methane leakage points was constructed to simulate realistic leak scenarios. An infrared emitter–detector pair was installed at selected locations along the pipeline to monitor variations in received signal intensity. The detected analog signal is conditioned and digitized using a microcontroller, then transmitted to a cloud platform via a cellular communication module for remote monitoring.

Artificial intelligence techniques are implemented to analyze the infrared sensor data and distinguish between normal operating conditions and methane leakage events. Machine learning models are trained using labeled experimental data to classify leak presence and estimate leak severity based on signal patterns and temporal variations. A software application retrieves data from the cloud and provides real-time leak alerts, signal visualization, and event logging.

System performance is evaluated through controlled experiments by varying leak rates and operating conditions. Key performance metrics include detection accuracy, response time, false alarm rate, and communication reliability. The results demonstrate the feasibility of combining infrared sensing, cellular communication, and AI-based analysis for effective methane leak detection in pipeline systems.

Acknowledgments

The project team would like to express their sincere appreciation to their project advisor and supervising faculty for their guidance, technical support, and constructive feedback throughout the development of this project. Their expertise and direction were essential in shaping the project's scope, methodology, and implementation.

The project team also acknowledges the support provided by the laboratory staff and department facilities, which enabled the successful construction and testing of the experimental setup. Their assistance with equipment access and safety procedures is gratefully appreciated.

Finally, the project team would like to thank all individuals and organizations who provided technical advice, resources, or support during this work. Their contributions played an important role in the completion of the project.

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Table of Contents

Abstract iii
Acknowledgments iv
AI Declaration Statement v
Table of Contents vii
List of Figures ix
List of Tables ix
List of Abbreviations x
1.1 Overview 11
1.2 Introduction 11
1.3 Fundamental Scientific and Engineering Concepts 13
1.3.1 Infrared Absorption Principle for Methane Detection 13
1.3.2 Infrared Sensing Configuration 15
1.3.3 Signal Conditioning and Data Acquisition 15
1.3.4 Data Transmission and Cloud-Based Monitoring 16
1.3.5 Data-Driven Analysis and Artificial Intelligence 16
1.3.6 System-Level Integration 17
1.3.7 Summary 17
Chapter 2 18
Project Definition 18
2.1 Project Summary 18
2.1.1 Project Statement 18
2.1.2 Project Objectives Statement 18
2.1.3 Deliverables 18
2.1.4 Customer Needs 19
2.1.5 Standards 19
2.1.6 Constraints 20
2.1.7 Specifications 21
2.2 Detailed Project Definition Discussion 23
2.2.1 Discussion of Deliverables 23
2.2.2 Stakeholder Analysis 24
2.2.3 Discussion of Constraints 25
2.2.4 Discussion of Specifications and Their Thorough Justifications 26
2.2.5 QFD Discussion 27
2.3 Project Impact 28
2.4 Discipline-Specific Contributions 32

2.4.1 Team Management 34
Chapter 3.....35
Design and Development.....35
3.1 Design Concept.....36
3.2 Concept Selection 38
3.3 Concept Evaluation.....41
3.4 Conceptual Prototype Description.....46
Chapter 4.....48
Final Design.....48
4.1 Overall Description.....48
4.2 Detailed Design Description.....50
4.2.1 System Overview (EE Part).....50
4.2.2 System Overview (PETE part) 56
4.2.3 System Overview (CS part).....58
4.3 Final Design Analysis and Results 65
4.3.1 Final Design Analysis (EE Part).....65
4.3.2 Final Design Analysis (PETE Part) 70
4.3.3 Final Design Analysis (CS Part).....73
4.4 Bill of Material 84
4.5 Cost Analysis 84
4.6 Engineering Drawing.....86
Chapter 5.....86
Conclusions and Recommendations 86
5.1 Conclusion 86
5.2 Recommendations.....87
References.....89
Appendix C.....92
Appendix E Point-By-Point Reply to Scope Form and CDR Presentation Reviewers' Comments 93
Appendix F Point-By-Point Reply to CDR Report Reviewers' Comments 103
Appendix G Point-By-Point Reply to Final Design Presentation Reviewers' Comments 109

List of Figures

Figure 1: Beer–Lambert Infrared Absorption Principle for Methane Detection 12
Figure 2: Beer–Lambert Law Representation and Parameter Definitions 12
Figure 3: Project QFD 26
Figure 4: Multidisciplinary Prototype Summary 31
Figure 5: Design Concept Fan 37
Figure 6: Radar Chart Evaluation of Four Detection Methods 43
Figure 7: Conceptual Infrared Methane Leak Detection Layout 45
Figure 8: Infrared Methane Detection Block Diagram 45
Figure 9: Multidisciplinary Block Diagram 48
Figure 10: EE Detailed Block Diagram of the Methane Detection System 48
Figure 11: Laboratory Test Validation for Methane Leak Detection 55
Figure 12: Backend Architecture & Data Flow 57
Figure 13: Final Engineering Drawing (Final Assembly) 84

List of Tables

Table 1: Project Constraints and Specifications 20
Table 2: Discipline Specific Courses 31
Table 3: Team Management 32
Table 4: SCAMPER Analysis 35
Table 5: Selection Criteria and Weighting Factors for Concept Evaluation 41
Table 6: Concept 1 – Infrared Methane Leak Detection 42
Table 7: Concept 2 – Distributed Fixed Sensors 43
Table 8: Concept 3 – Rail-Guided Mobile Robot 44
Table 9: Concept 4 – Portable Handheld Inspection Unit 45
Table 10: Pugh Matrix (Radar Chart Input) 46
Table 11: Methane Leak Severity Classification Based on LEL 76
Table 12: Summary of Project Constraints and Specifications Attainment in the Final Design 78
Table 13: Bill of Material 82

List of Abbreviations

ADC Analog-to-Digital Converter

AI Artificial Intelligence

AES Advanced Encryption Standard

CH_4 Methane Gas

CO_2 Carbon Dioxide

CS Computer Science

EE Electrical Engineering

IR Infrared

LEL Lower Explosive Limit

MCU Microcontroller Unit

NDIR Non-Dispersive Infrared

NEP Noise Equivalent Power

OSHA Occupational Safety and Health Administration

PETE Petroleum Engineering

PV Photovoltaic

QFD Quality Function Deployment

Rx Receiver

Tx Transmitter

UI User Interface

ADC Analog-to-Digital Converter

TLS Transport Layer Security

Chapter 1
Background

1.1 Overview

Methane leakage from pipeline systems represents a persistent challenge in terms of safety, environmental impact, and operational efficiency. Pipelines often extend across long distances and operate in locations that are difficult to continuously monitor, making early leak detection a complex task. Undetected methane leaks can escalate into hazardous situations, contribute to greenhouse gas emissions, and result in increased operational and maintenance costs.

Current methane detection practices rely heavily on fixed sensing units or periodic inspections, which may provide limited coverage and delayed response. These approaches can struggle to detect small or intermittent leaks and are often affected by environmental conditions and sensor noise. As a result, there is a need for alternative detection approaches that allow controlled evaluation of sensing performance and system behavior.

This design project addresses these challenges by offering a bench-top methane leak detection system that enables controlled laboratory testing of methane sensing concepts. By focusing on a laboratory-scale prototype, the project provides a safe, repeatable, and cost-effective platform to study methane detection using infrared absorption principles and integrated system components.

1.2 Introduction

Methane is widely used and transported in oil and gas infrastructure, making reliable leak detection a critical requirement for safe operation. Even minor methane leaks can pose significant safety risks due to the flammable nature of the gas and can contribute to environmental harm through increased greenhouse gas emissions. The difficulty of continuously monitoring extensive pipeline networks highlights the need for detection systems that can identify leaks early and reliably.

The motivation for this project arises from the limitations of existing methane detection approaches, particularly in scenarios where controlled testing and system validation are required. Many commercial detection systems are designed for fixed installations or field deployment and are often costly, complex, and difficult to adapt for experimental evaluation. As a result, there is a gap between advanced industrial solutions and laboratory-scale systems that allow systematic investigation of methane sensing performance.

The oil and gas industry is the primary sector affected by methane leakage and stands to benefit from improved detection technologies. Pipeline operators, inspection of service providers, and safety and regulatory bodies all require dependable methods to identify leaks and assess system integrity. In recent years, the industry has increasingly adopted data-driven and intelligent monitoring approaches to enhance detection of reliability, reduce false alarms, and support operational decision-making. These approaches commonly combine sensing hardware with software-based data analysis, rather than relying solely on fixed threshold alarms.

Several methane detection solutions are currently available, including fixed gas detectors, portable handheld sensors, and advanced optical systems such as laser-based and remote sensing technologies. Fixed detectors provide continuous monitoring at specific locations but offer limited coverage along extended pipeline segments. Handheld sensors require manual inspection and may fail to detect transient or low-level leaks consistently. While advanced optical systems offer high sensitivity, they are typically expensive and impractical for laboratory-scale experimentation or early-stage concept validation. In addition, many existing systems rely on simple threshold-based triggering, which can be sensitive to noise and environmental variability.

To address these limitations, the proposed project integrates artificial intelligence as a supportive analysis layer within a bench-top methane detection system. Rather than replacing the physical sensing mechanism, AI techniques are used to analyze infrared sensor data patterns and assist in distinguishing normal operating conditions from methane leakage events. This data-driven approach aims to improve detection reliability and reduce false alarms caused by transient disturbances or sensor noise. By combining infrared sensing, embedded processing, cloud-based data handling, and AI-assisted analysis, the project aligns with current industry trends toward intelligent monitoring systems while maintaining a controlled and experimentally verifiable laboratory scope.

1.3 Fundamental Scientific and Engineering Concepts

This project is based on a combination of fundamental scientific principles and core engineering concepts drawn from petroleum engineering, electrical engineering, and computer science. These concepts collectively enable the detection of methane leaks, the conversion of physical phenomena into measurable electrical signals, and the analysis of sensor data within a controlled laboratory environment.

1.3.1 Infrared Absorption Principle for Methane Detection

The fundamental scientific principle underlying the proposed methane detection system is infrared (IR) absorption. Methane molecules absorb infrared radiation at specific wavelengths due to molecular vibrational modes. When infrared light propagates through a region containing methane gas, a portion of the incident radiation is absorbed, resulting in a reduction in the transmitted light intensity.

This absorption behavior is described by the Beer–Lambert principle, which relates the amount of absorbed radiation to the gas concentration and the optical path length. The conceptual behavior of this relationship is illustrated in Figure 1, where infrared radiation emitted from a light source pass through a gas-filled optical path of length L, and the transmitted light intensity decreases due to absorption by methane molecules. The mathematical representation and parameter definitions of this relationship are shown in Figure 2, where the absorbance depends on the molar absorption coefficient, gas concentration, and optical path length.

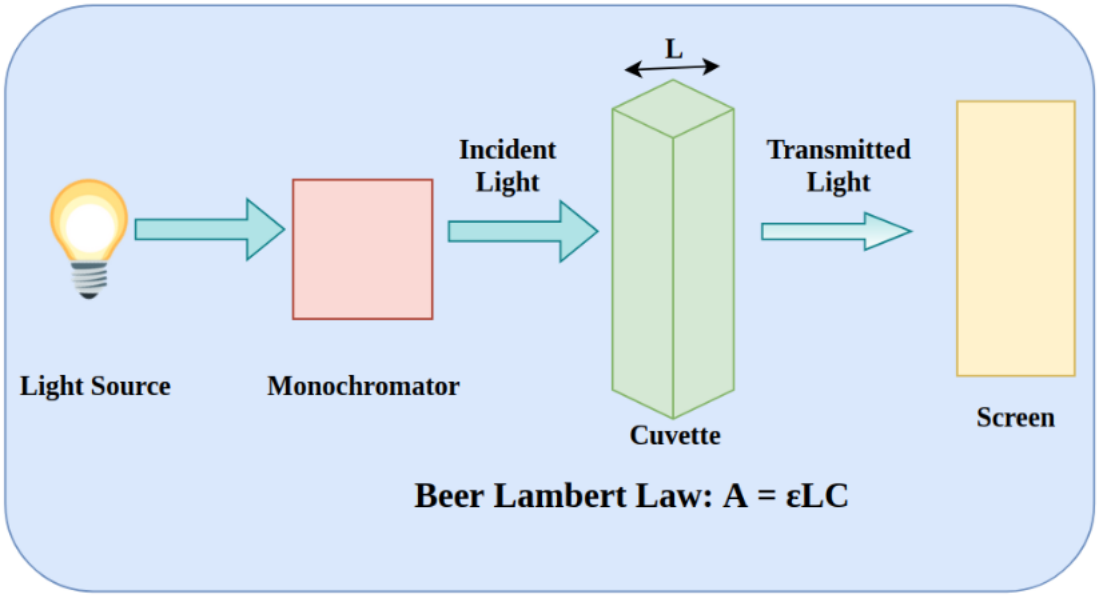

Figure 1: Beer–Lambert infrared absorption principle for methane detection

$$A = \varepsilon c l$$

A	Absorbance	
ε	Molar absorption coefficient	$M^{-1}cm^{-1}$
c	Molar concentration	M
l	optical path length	cm

Figure 2: Beer–Lambert law representation and parameter definitions

In the context of this project, the Beer–Lambert principle is not used to calculate precise methane concentration values. Instead, it provides a physical basis for understanding how the presence of methane causes measurable attenuation of infrared radiation. The system relies on detecting relative changes in transmitted signal intensity, which is sufficient for identifying leak events under controlled laboratory conditions.

From a petroleum engineering perspective, this principle supports the analysis of methane behavior in pipeline leak scenarios, where gas presence within the optical path indicates

leakage. Controlled leak simulation allows systematic evaluation of how methane accumulation affects optical transmission and detection reliability.

1.3.2 Infrared Sensing Configuration

To implement the infrared absorption principle, the system uses an infrared emitter–detector pair positioned across selected regions of the pipeline test section. The emitter generates continuous infrared radiation directed toward the detector. Under normal operating conditions, the detector receives a stable signal corresponding to unobstructed transmission.

When methane leaks into the optical path, partial absorption of infrared radiation occurs, leading to a reduction in the detected signal intensity. This configuration represents a non-dispersive infrared sensing approach, which is suitable for bench-top experimentation due to its simplicity, robustness, and repeatability. Mechanical alignment, optical shielding, and fixed geometry are used to minimize the influence of ambient light and environmental disturbances.

1.3.3 Signal Conditioning and Data Acquisition

From an electrical engineering standpoint, the infrared detector produces a low-amplitude analog electrical signal that is susceptible to noise and interference. To ensure reliable measurement, signal conditioning is applied before digital processing. This includes amplification to increase signal magnitude and filtering to suppress unwanted noise components.

The conditioned signal is then digitized using an analog-to-digital converter (ADC) integrated within a microcontroller. Digitization enables consistent sampling, numerical representation of the sensor output, and time-stamping of measurements. These steps are essential for reliable data transmission and subsequent software-based analysis.

In the final system design, digitized measurements are structured into time-stamped data frames and transmitted securely to the backend using HTTPS, ensuring encrypted communication between the embedded system and cloud infrastructure.

The microcontroller functions as the central embedded system, managing sensor sampling, local data buffering for time-series continuity, and secure communication with external platforms.

1.3.4 Data Transmission and Cloud-Based Monitoring

Communication engineering concepts are applied through secure HTTPS-based data transmission from the embedded system to the backend cloud platform. All transmitted data is secured using Transport Layer Security (TLS), which performs authenticated key exchange and encrypted communication using AES-based symmetric encryption after session establishment. This architecture reflects modern monitoring system designs used in industrial and research applications.

The FastAPI backend and cloud database serves as an intermediate layer that allows sensor data to be accessed by higher-level software applications without direct physical interaction with the hardware.

1.3.5 Data-Driven Analysis and Artificial Intelligence

From a computer science perspective, the project employs data-driven analysis techniques to support methane leak detection. Rather than relying solely on fixed threshold values, sensor data is analyzed to identify patterns associated with normal operation and leak events. Features such as signal magnitude variation and temporal behavior are extracted from the infrared measurements.

An unsupervised anomaly detection approach based on the Isolation Forest algorithm is applied to model normal operating behavior and detect deviations indicative of methane leakage. Instead of relying on labeled leak data, the model learns baseline signal patterns and assigns an anomaly score to new observations based on structural deviation in the feature space. The role of artificial intelligence in this project is to provide predictive anomaly scoring and

graded decision support, complementing the physical sensing mechanism rather than replacing it.

1.3.6 System-Level Integration

The integration of infrared sensing, signal conditioning, embedded processing, wireless communication, and data-driven analysis forms a complete methane leak detection system. Each subsystem performs a specific function, and overall performance depends on their coordinated operation. Real-time inference within the backend ensures that anomaly detection results are generated within the defined system response constraint, while secure transmission and structured data handling maintain reliability across subsystem boundaries. From a systems engineering perspective, the project emphasizes integration and experimental validation rather than component-level optimization.

By combining petroleum engineering context, electrical sensing hardware, and computer science–based data analysis, the project provides a multidisciplinary platform for evaluating methane leak detection behavior, response time, and reliability under controlled laboratory conditions.

1.3.7 Summary

The application of infrared absorption physics, basic signal processing, embedded system design, wireless communication, and data-driven predictive anomaly detection forms the scientific and engineering foundation of the proposed methane detection system. These concepts collectively enable the implementation and evaluation of a bench-top prototype that supports controlled testing, performance assessment, and future scalability while remaining aligned with academic design requirements and project scope.

Chapter 2

Project Definition

This chapter defines the proposed methane leak detection project by outlining its scope, requirements, and design boundaries. Section 2.1 presents a concise project summary, including the project statement, objectives, deliverables, customer needs, applicable standards, constraints, and technical specifications. Section 2.2 provides a detailed discussion and justification of these elements to ensure clear alignment between customer requirements, system design decisions, and measurable performance targets. Section 2.3 describes the expected project impact, highlighting the safety, environmental, and operational value of the proposed system and its relevance to industry needs. Section 2.4 outlines the discipline-specific contributions and project management approach, clarifying the responsibilities of PETE, EE, and CS team members and how their work integrates to deliver the complete system. Together, these sections establish a structured foundation for the subsequent design and development phases of the project.

2.1 Project Summary

2.1.1 Project Statement

The project aims to design and implement a bench-top methane leak detection system based on infrared absorption and AI anomaly to enable reliable detection and evaluation of methane leaks under controlled laboratory conditions.

2.1.2 Project Objectives Statement

This project aims to develop a laboratory-scale methane detection system that senses methane-induced infrared signal attenuation, processes and transmits the data to a cloud platform, and presents leak information through a software application for analysis and evaluation.

2.1.3 Deliverables

The deliverables of this project include the following:

1. A benchtop pipeline test rig with controlled methane leak simulation.
2. An infrared emitter–detector sensing subsystem integrated with signal conditioning and a microcontroller.
3. A wireless communication and cloud data pipeline for remote data transmission and storage.
4. A software application that displays methane detection status, signal visualization, and event logs.
5. Documented system testing results in demonstrating detection performance under controlled laboratory conditions.

2.1.4 Customer Needs

The customer needs identified for this project are as follows:

1. The system must detect methane leaks reliably under controlled conditions.
2. The system must support safe laboratory testing without requiring field deployment.
3. The system must provide clear and interpretable leak indication to the user.
4. The system must reduce false alarms caused by noise or transient effects.
5. The system must support future scalability and adaptation to more advanced monitoring solutions.

2.1.5 Standards

The design and operation of the proposed methane leak detection system follow a limited set of safety, laboratory, communication, and data security standards that are directly relevant to the project constraints and specifications. These standards are considered to ensure safe laboratory testing, reliable system operation, and secure data transmission:

- OSHA (Occupational Safety and Health Administration) standards related to the handling of flammable gases and safe operation of electrical equipment during laboratory testing.
- Lower Explosive Limit (LEL) safety guidelines for methane, used to define safe detection thresholds below hazardous concentration levels.
- Advanced Encryption Standard (AES-128) for securing transmitted sensor data.
- KFUPM laboratory safety regulations governing calibration of gases, experimental procedures, and laboratory operation.

2.1.6 Constraints

The proposed methane leak detection system is subject to the following constraints, which define the boundaries and limitations within which the design, implementation, and testing must be conducted:

1. The total cost of the prototype shall not exceed 15,000 SAR.
2. System sensitivity shall be limited to ppm-level methane detection, prioritizing low power consumption and system simplicity over ppb-level sensitivity.
3. Methane measurement accuracy may be affected by environmental factors, including humidity, carbon dioxide (CO_2), and other hydrocarbons.
4. Optical alignment between the infrared source and detector must be maintained to ensure reliable methane detection.
5. The system shall operate under outdoor pipeline conditions, including 0–60 °C temperature, up to 95% relative humidity (non-condensing), and dust exposure.
6. Calibration of gases and testing procedures must comply with KFUPM laboratory requirements.
7. System operation must comply with OSHA safety standards, particularly those related to flammable gas handling and laboratory electrical safety.
8. Infrared source of intensity drifts, and ambient infrared noise may affect long-term signal stability.

9. Leak detection time depends on the methane plume reaching the optical sensing path, which may introduce variability in detection response.
10. Cellular network availability and latency may affect the timing of data transmission and application notifications.

2.1.7 Specifications

The proposed methane leak detection system shall satisfy the following technical and functional specifications. These specifications are stated as measurable and verifiable performance requirements and are directly derived from the project constraints and objectives:

1. The system shall detect methane (CH_4) only as the target gas.
2. Methane detection shall be based on open-path infrared absorption using a fixed emitter–detector (NDIR / opto-pair) configuration.
3. The infrared source shall operate at methane absorption wavelengths within 1.63–1.69 µm (near-infrared) or 3.2–3.4 µm (mid-infrared).
4. The optical path length between the infrared emitter and detector shall be fixed and known during operation.
5. The methane detection threshold shall be set below 1% of the Lower Explosive Limit (LEL).
6. Local methane leak detection at the microcontroller level shall occur within ≤ 2 seconds of sensing.
7. End-to-end alert delivery to the user application shall occur within ≤ 10 seconds.
8. The infrared signal shall be sampled at a rate of ≥ 10 Hz.
9. Narrowband optical filtering shall be used to isolate methane-sensitive wavelengths with a defined passband matching the selected absorption range.
10. AI-based leak classification shall achieve a minimum accuracy of ≥ 70% in distinguishing true leak events from noise.
11. Methane monitoring shall be performed at fixed sensing locations along the pipeline.
12. Sensor data shall be transmitted to the cloud using cellular communication.

13. Data transmission reliability shall maintain a packet loss rate of < 1%.
14. The system shall provide methane leak alerts and system status information through a mobile and/or desktop application.
15. Total system power consumption shall not exceed 5 W during operation.
16. The system shall support at least 6 hours of continuous operation using battery power.
17. All transmitted data shall be secured using AES-128 encryption.

Table 1: Project Constraints and Specifications

Constraints and Specifications	Off-the-Shelf	Project Specific (Not Off-the-Shelf)	Responsible Department		
			EE	PETE	CS
Constraint 1		✓	✓	✓	✓
Constraint 2		✓	✓	✓	
Constraint 3		✓	✓	✓	✓
Constraint 4		✓	✓		
Constraint 5		✓	✓	✓	
Constraint 6		✓	✓	✓	✓
Constraint 7		✓	✓	✓	
Constraint 8		✓	✓		✓
Constraint 9		✓		✓	
Constraint 10		✓	✓		✓
Specification 1		✓	✓	✓	
Specification 2	✓		✓	✓	
Specification 3	✓		✓	✓	
Specification 4		✓	✓	✓	
Specification 5		✓	✓	✓	

Specification 6		✓	✓		
Specification 7		✓	✓		✓
Specification 8		✓	✓		
Specification 9	✓		✓		
Specification 10		✓			✓
Specification 11	✓		✓	✓	
Specification 12	✓		✓		✓
Specification 13		✓			✓
Specification 14		✓			✓
Specification 15		✓	✓		
Specification 16		✓	✓		
Specification 17		✓			✓
Integrated Specification 1		✓	✓		✓
Integrated Specification 2		✓	✓		✓
Integrated Specification 3		✓			✓

2.2 Detailed Project Definition Discussion

2.2.1 Discussion of Deliverables

The primary deliverables of this project are designed to demonstrate the feasibility of integrating an infrared-based methane sensing system with data-driven analysis for early leak detection. The deliverables focus on developing a functional prototype, generating experimentally validated data, and providing a clear evaluation of system performance within a controlled laboratory environment. The first major deliverable is a functional infrared methane sensing prototype based on an open-path infrared absorption principle. This prototype

includes an infrared source, an optical detector, signal conditioning circuitry, and a microcontroller for data acquisition. The sensing unit is capable of continuously measuring methane concentration in parts per million (ppm) and transmitting these measurements for further processing. This deliverable validates hardware integration and ensures reliable methane detection under laboratory conditions. The second deliverable is a time-series methane dataset obtained through controlled laboratory experiments. Methane leak scenarios, including no-leak, slow-leak, and sudden-leak conditions, are simulated in a safe and controlled manner. The infrared sensor records methane concentration at fixed time intervals, producing structured datasets with timestamps. These datasets serve as the foundation for subsequent data analysis and are essential for evaluating detection performance beyond traditional threshold-based approaches. The third deliverable is an AI-based leak detection and classification module developed using the collected sensor data. This module processes methane concentration trends over time to distinguish between normal background variations and abnormal leak patterns. The AI component is designed to enhance early detection capability while operating alongside conventional safety thresholds, ensuring that system reliability and safety are preserved. Performance metrics such as detection accuracy and response time are used to assess the effectiveness of this deliverable. The fourth deliverable is a real-time monitoring and alert output, which presents sensor readings and AI-based decisions to the user. This includes live visualization of methane concentration, system status indicators, and warning notifications when abnormal behavior is detected. The output demonstrates how the proposed system can support operators in identifying potential leaks earlier and more reliably than traditional fixed-threshold systems. Collectively, these deliverables demonstrate a complete and integrated system that combines infrared methane sensing with intelligent data interpretation, fulfilling the project objectives.

2.2.2 Stakeholder Analysis

2.2.2.1 Stakeholder list and prioritization

The primary stakeholder of this project is oil and gas operating companies responsible for pipeline safety and methane monitoring. These organizations are the direct users of the proposed system and have the highest influence on design requirements related to detection accuracy, reliability, and response time. Secondary stakeholders include safety and regulatory authorities, whose interest lies in ensuring compliance with occupational and environmental standards. Maintenance and inspection personnel are also key stakeholders, as they rely on clear

and reliable system outputs to support timely inspection and repair activities. Additional indirect stakeholders include environmental agencies and nearby communities, who are affected by methane emissions and overall pipeline safety. Academic supervisors and project sponsors are also stakeholders, influencing project scope, feasibility, and adherence to academic and ethical standards.

2.2.2.2 Customer Needs and Requirements

The customer for this project is the oil and gas operating company, which requires early and reliable methane leak detection to enhance safety and reduce operational risks. Accurate and stable methane measurements are essential to minimize false alarms and ensure dependable monitoring under normal operating conditions. The customer also requires timely alerts and clear system outputs to support fast decision-making during potential leak events. Operational simplicity, low maintenance, and compliance with safety and regulatory standards are additional key requirements. These customer needs directly guide the system specifications and design choices discussed in later sections.

2.2.3 Discussion of Constraints

The design and implementation of the proposed infrared-based methane detection system are subject to several constraints that influence system performance, feasibility, and scope. These constraints were carefully considered to ensure that the project remains realistic and achievable within undergraduate laboratory and resource limitations. One of the primary constraints is budget, as the total prototype cost must remain within the allocated limit (15000 SAR). This constraint affects component selection, sensor type, and communication hardware, requiring the use of commercially available infrared sensing modules rather than custom-built optical systems. Time constraints also play a significant role, as the project must be completed within a single academic term. This limits the complexity of system development, experimental testing, and data collection, and necessitates a focused and well-defined project scope. There are also technical and physical constraints associated with infrared methane sensing. Measurement accuracy depends on maintaining proper optical alignment and sufficient methane exposure within the sensing path. Environmental factors such as humidity, ambient infrared noise, and airflow may influence sensor readings and introduce variability in the collected data. Safety and regulatory constraints further affect the project's design. All methane testing must be conducted at safe concentrations and in compliance with laboratory safety regulations and occupational safety standards. These constraints limit the maximum methane

levels that can be tested and require controlled experimental procedures. Finally, communication and power constraints influence system operation, as data transmission reliability and power consumption must remain within specified limits. Network availability and latency may affect alert delivery timing, which is considered in system performance evaluation. Overall, these constraints guide the design decisions and ensure that the proposed system balances performance, safety, cost, and feasibility while meeting project objectives.

2.2.4 Discussion of Specifications and Their Thorough Justifications

The project specifications were defined to ensure that the proposed infrared-based methane detection system meets customer needs while remaining feasible within the identified constraints. Each specification is measurable and directly supports the project objectives of early leak detection, reliability, and safe operation. The system is specified to detect methane (CH_4) only, as methane is the primary gas of concern in pipeline leakage scenarios. Detection is based on open-path infrared absorption (NDIR) at methane-specific wavelengths, which is justified by the need for high selectivity, low drift, and long-term stability compared to non-optical sensing methods. A detection threshold below 1% of the Lower Explosive Limit (LEL) is specified to enable early warning well before hazardous conditions occur. This specification directly addresses the customer's requirement for enhanced safety and proactive leak identification. To support timely response, the system is required to perform local leak detection within two seconds and deliver alerts to the user within a defined response time, ensuring actionable information is available promptly. The sampling rate of the infrared detector is specified to be sufficient to capture transient methane leaks, which is essential for distinguishing real leak events from background fluctuations. Additionally, narrowband optical filtering is required to isolate methane-sensitive wavelengths and reduce interference from ambient infrared sources, improving measurement accuracy. Power-related specifications, including low total power consumption and battery-supported operation, are justified by the need for continuous monitoring without frequent maintenance or power interruption. Communication specifications, such as reliable data transmission and secure data handling, support the customer's requirement for dependable system operation and data integrity. Overall, the selected specifications balance performance, safety, and practicality. They are directly derived from customer needs and project constraints and ensure that the system achieves its intended functionality.

2.2.5 QFD Discussion

The Quality Function Deployment (QFD) was used to systematically translate customer needs (WHATs) into measurable engineering specifications (HOWs) for the proposed fixed infrared methane detection system with AI-based anomaly detection. High-priority customer needs such as safety, time efficiency, and minimal false alarms show strong relationships with technical specifications including methane detection threshold, alert response time, end-to-end alert delivery, and AI classification accuracy, reflecting the importance of early and reliable leak detection. Ease of monitoring is strongly linked to user interface clarity and supported by reliable communication and timely alert delivery. Several meaningful trade-offs were identified in the roof correlation of the QFD. Improving alert response time and wireless reliability can increase power consumption, which negatively affects battery operating time, requiring a balance between fast detection and long-term operation. Increasing data security through encryption improves data integrity but may introduce minor delays in end-to-end alert delivery. A strong positive synergy exists between power consumption and battery life, while improved optical wavelength selectivity positively supports AI classification accuracy by enhancing signal quality. Competitor analysis shows that traditional non-optical fixed gas detectors perform well in basic safety and continuous monitoring but score lower in early detection and false-alarm reduction, while conventional infrared detectors provide stable measurements but rely on threshold-based alarms without intelligent analysis. In contrast, the proposed system combines infrared sensing with AI to improve early detection and decision support. Target and threshold values, such as detection below 1% LEL, alert delivery within seconds, and AI accuracy ≥70%, were selected to balance safety requirements, performance, and feasibility within prototype constraints rather than full industrial optimization.

User inteface clarity															
Data Security level															
Wirless reliability															
Power consumption											–	–			
Battery operating time										++					
CH4 adsorption wavelen															
AI classification accuracy									+						
End to end alert delivery											++	–			
Alert response time							+			–	–				
Methane detection thre															

What	Who: Oil & Gas Pipeline Operating Companies	Who: Pipeline Maintenance Contractors	Who: HSE Service Providers	How: Methane detection threshold ↓	How: Alert response time ↓	How: End to end alert delivery ↓	How: AI classification accuracy ↑	How: CH4 adsorption wavelength ↑	How: Battery operating time ↑	How: Power consumption ↓	How: Wirless reliability ↑	How: Data Security level ↑	How: User inteface clarity ↑	Now: N -Non Optical Gas Detectors (Drager) T-Traditional IR Methane Detectors (No AI), (Honeywell) 1	2	3	4	5
				%	s	s	%	μm	ns	W	%	STD	qlt					
Ease of monitoring	6	5	7	△	○	○	△				○	△	◉			N T		
Time Efficient	5	8	4	△	◉	○	△	○			○					N	T	
Minimal false alarms	6	4	10	○	△	◉	◉				△				N	T		
Ease of installation & deployment at fixed points	4	9	3					◉	○	△						T	N	
Coverage at critical pipeline segments	9	6	3	◉	○	◉	○	◉	○							N	T	
Safety	7	4	10	◉	○	◉	◉	△			◉	◉					N	T
Importance (Oil & Gas Pipeline Operating Companies)				16%	11%	21%	14%	13%	4%	0%	9%	6%	5%					
Importance (Pipeline Maintenance Contractors)				13%	14%	19%	12%	18%	5%	1%	9%	5%	5%					
Importance (HSE Service Providers)				14%	10%	22%	18%	7%	2%	0%	12%	9%	6%					
Target Value				o.5	<-1s	<-8s	->80	1.63-	>-8	<-4	>-99.5	AES 256	100					
Threshold Value				1	<-2s	<-10	->70	1.63	>-6	<-5	>-99	'128	80					

Figure **3**: Project QFD

2.3 Project Impact

Economic Perspective:

The proposed infrared-based methane detection system has a positive economic impact by supporting early identification of methane leaks and reducing financial losses associated with undetected gas emissions. Methane leaks can result in direct product loss, increased operational costs, equipment damage, and unplanned shutdowns. In severe cases, delayed detection may lead to costly emergency response actions, environmental penalties, and extended downtime. By enabling

earlier detection through stable infrared sensing and intelligent data analysis, the proposed system allows operators to intervene before leaks escalate into major failures, thereby reducing repair, recovery, and shutdown-related costs. From a maintenance perspective, early identification of abnormal methane behavior supports a shift from reactive maintenance to condition-based maintenance. Instead of responding only after alarms or failures occur, operators can plan inspections and corrective actions based on observed trends in methane concentration. This approach improves maintenance scheduling, reduces unnecessary site visits, and minimizes labor and operational disruptions. Early intervention also helps prevent secondary damage to pipeline components, valves, and surrounding equipment, which can significantly increase repair costs if leaks are left unaddressed. In addition, the use of infrared sensing technology improves measurement stability and long-term reliability by minimizing sensor drift and poisoning effects commonly associated with non-optical sensors. Improved reliability reduces false alarms and unnecessary shutdowns, which are often expensive and disruptive to operations. More accurate detection also ensures that maintenance resources are directed toward actual leak events rather than false positives. Overall, the proposed system contributes to improved asset integrity management, reduced methane losses, and lower operational and maintenance costs, resulting in measurable economic benefits for oil and gas operators.

Societal Perspective:

The proposed infrared-based methane detection system has a positive societal impact by enhancing safety in environments where methane pipelines and gas facilities operate, particularly in proximity to workers and surrounding communities. Methane leaks pose serious hazards, including fire and explosion risks, which can lead to injuries, loss of life, and damage to nearby infrastructure. By supporting early identification of abnormal methane behavior, the proposed system reduces the likelihood of such incidents and contributes to safer operating conditions. For workers and maintenance personnel, reliable methane monitoring improves situational awareness and reduces exposure to hazardous conditions. Early warnings allow operators to take preventive actions before methane concentrations reach dangerous levels, thereby lowering the risk of emergency situations. Clear system outputs and alerts also support faster and more informed decision-making, which is critical during abnormal operating conditions. This contributes to safer work

environments and helps reduce occupational health and safety risks. At a broader societal level, improved methane detection supports public confidence in energy infrastructure and industrial operations. Communities located near pipelines or processing facilities are increasingly concerned about safety and environmental risks associated with energy production. Demonstrating the use of reliable monitoring technologies helps address these concerns and promotes trust between operators and the public. Furthermore, the project aligns with societal expectations for responsible engineering practices that prioritize safety, risk prevention, and transparency. By focusing on early detection rather than emergency response alone, the proposed system supports proactive safety management. This approach contributes to the overall well-being of workers, nearby communities, and emergency response teams by reducing the likelihood and severity of hazardous incidents.

Environmental Perspective:

Methane is a potent greenhouse gas with a significantly higher global warming potential than carbon dioxide over shorter time periods. As a result, uncontrolled methane leaks contribute substantially to climate change and environmental degradation. The proposed infrared-based methane detection system has a positive environmental impact by enabling earlier identification of leaks and reducing unnecessary methane emissions into the atmosphere. Infrared sensing technology provides accurate and selective methane measurement, allowing small concentration changes to be detected before leaks become large or persistent. When combined with data-driven analysis, the system improves the likelihood of identifying abnormal emission patterns that may not trigger traditional threshold-based alarms. Early detection enables faster intervention, reducing the duration and magnitude of methane release and limiting its environmental impact. Although the project is implemented at a laboratory scale, it demonstrates principles that are directly applicable to real-world emission reduction efforts. Improved monitoring supports better methane management practices, which are increasingly emphasized by environmental regulations and global initiatives aimed at reducing greenhouse gas emissions from the oil and gas sectors. By minimizing methane losses, the system contributes to more sustainable energy production and improved environmental performance. In addition, reducing methane emissions has secondary environmental benefits, such as improved air quality and reduced formation of ground-level ozone. Overall, the proposed system highlights the environmental advantages of integrating

advanced sensing technologies with intelligent data analysis and supports broader efforts to mitigate climate-related impacts associated with methane emissions.

Ethical Concerns (Multidisciplinary Perspective):

From the Petroleum Engineering (PETE) perspective, the primary ethical concern is ensuring the safety of personnel, facilities, and surrounding communities by minimizing the risks associated with methane leaks. Methane is a flammable gas and potent greenhouse gas, and failure to detect leaks early can lead to explosions, environmental harm, and regulatory violations. PETE ethical responsibility focuses on safe operations, environmental stewardship, and prevention of hydrocarbon losses. From the Electrical Engineering (EE) perspective, ethical concerns center on system reliability, sensor accuracy, and safe hardware operation. Incorrect electrical design, poor signal integrity, or unreliable power systems could lead to false alarms or missed detections, potentially endangering lives. EE ethics require ensuring robust hardware integration, safe electrical practices, and dependable system performance. From the Computer Science (CS) perspective, ethical concerns involve responsible use of data-driven analysis and automation. Over-reliance on AI outputs without transparency or human oversight may lead to unsafe decisions. There is also an ethical obligation to ensure data integrity, avoid misleading classifications, and clearly communicate system limitations.

NSPE Code of Ethics Application:

The project adheres to the NSPE Code of Ethics, particularly the principle that engineers must "hold paramount the safety, health, and welfare of the public."

- PETE contributes by prioritizing leak prevention and environmental protection.
- EE ensures safe and reliable sensor operation and electrical design.
- CS ensures that data processing and AI models are used responsibly and transparently.

The team also follows the NSPE requirement to perform work only within areas of competence and to present results honestly. The system is clearly identified as a laboratory-scale prototype and not an industrially certified safety system.

Stakeholders and Ranking:

The highest-priority stakeholders are workers and operators, whose safety depends on reliable methane detection. Second are nearby communities that are affected by potential leaks or accidents. Third are oil and gas operating companies, responsible for safe

operations and compliance. Additional stakeholders include regulatory authorities, environmental agencies, and maintenance personnel, who rely on accurate data for enforcement and decision-making.

Proposed Ethical Action and Resolution:

To address these ethical concerns, the system is designed as a decision-support tool, not a replacement for existing safety systems. PETE ensures conservative safety thresholds remain active, EE ensures hardware reliability and electrical safety, and CS ensures AI outputs are interpretable and used with human oversight. All testing is conducted under safe laboratory conditions, and system limitations are clearly documented to ensure ethical and professional responsibility across all disciplines.

2.4 Discipline-Specific Contributions

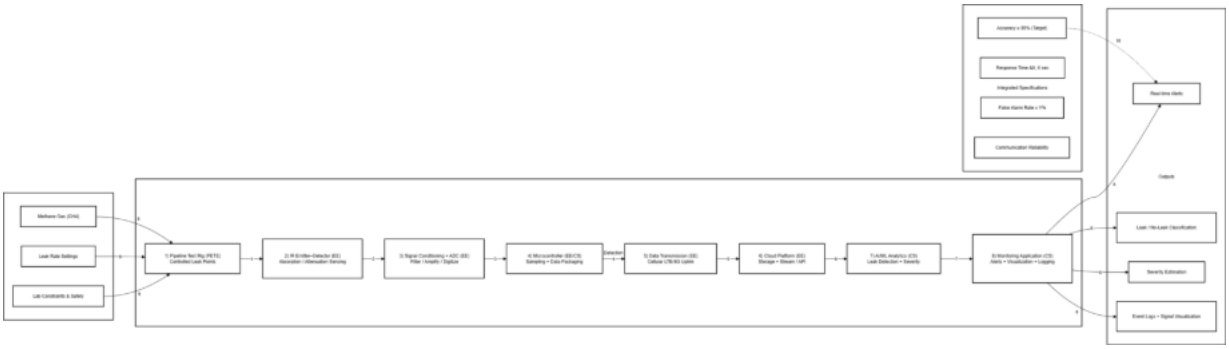

Figure **4**: Multidisciplinary Prototype Summary

Table 2: Discipline Specific Courses

No.	Discipline	Courses	Application Parts: Blocks (mandatory) and arrows (if applicable)
1.	**CS**	SWE463: Mobile Application Development	Monitoring Application → Real-time alerts, signal visualization, and event logging
2.	**PETE**	CHE204, PETE202 PETE403, CHEM101	Pipeline & Leak Modeling → Methane Release Scenarios → Gas Concentration Profiles → Sensor Calibration Inputs
3.	**EE**	EE 203 & 303 , EE 340, EE 370, EE 400	IR Transmitter (Tx) → Optical Path → IR Receiver (Photodiode) → Signal Conditioning (TIA & Filtering) → ADC (Microcontroller)
4.	**EE**	EE 203 & 303 , EE 340, EE 370, EE 400	Microcontroller → Local Processing → Cellular Communication Module → Cloud Data Uplink
5.	**CS**	ICS474: Big Data Analytics	Cloud Platform → Data storage, stream processing, and historical analysis Cloud → AI/ML Analytics

6.	**CS**	ICS381: Principles of Artificial Intelligence	AI/ML Analytics→ Leak detection and severity estimation Cloud Platform → AI/ML

2.4.1 Team Management

Table 3: Team Management

No.	Student Name	Major	Main Tasks				
			Constraints	Specs.	Integrated Specs.	Blocks	Arrows
1.	Hala	EE	Sensor noise, stable power, analog drift	IR alignment & sensitivity, filtering design, ADC resolution	Signal integrity and measurement stability	2, 3, 4	1, 2, 3
2.	Dana	EE	Network latency, packet loss, power consumption	Cellular LTE/4G uplink, data packaging, cloud interface	Communication reliability	5, 6	4, 5, 6
3.	Naflah	PETE	Safety limits, realistic leak simulation, repeatability	Leak point selection, methane leak rate ranges, test protocol	Valid operating conditions for reliable detection	1	9, 1

4.	Ragha d	CS	Data quality, model latency	**AI:** Feature Extraction & Preprocessin g. **App:** data visualization logic	Stable leak classification input	7, 8	6, 7, 8
5.	Maram	CS	Overfitting, class imbalance	**AI:** model training & tuning. **App:** alert triggering rules	Accuracy ≥ 70%, false alarm rate ≤ Y%	7, 8	6, 7, 8
6.	Elaf	CS	UI responsiveness , synchronizatio n delay	**AI:** Model Evaluation & Testing. **App:** UI/UX & notification handling	Real-time alerts, usability	7, 8	6, 7, 8

Chapter 3

Design and Development

This chapter describes the design and development process of the proposed infrared-based methane leak detection system. It explains how initial design concepts were generated, evaluated, and refined to arrive at the final system architecture. Section 3.1 presents the design concept generation process, including internal brainstorming, external research, and expert

consultation. Section 3.2 describes the selection of the final concept based on feasibility, safety, performance, and system requirements.

3.1 Design Concept

- **Internal Search:**

The internal search phase focused on generating multiple design concepts through structured brainstorming sessions conducted by the project team. Initial ideas included contact-based methane sensors, mobile robotic inspection platforms, distributed fixed sensor networks, and optical sensing approaches. These concepts were evaluated qualitatively based on safety, reliability, complexity, and feasibility.

To guide the concept generation process in a systematic manner, the SCAMPER methodology was applied. This approach enabled the team to explore design alternatives by substituting, combining, adapting, modifying, eliminating, and reversing elements of existing methane detection systems. Through SCAMPER analysis, traditional contact-based gas sensors were substituted with non-contact infrared sensing, mechanical components such as mobile robots were eliminated to reduce complexity, and the sensing geometry was modified to provide circumferential coverage around the pipeline.

The results of the SCAMPER analysis are summarized in Table 4, which highlights how structured ideation led to the development of an infrared-based, non-intrusive methane leak detection concept.

Table 4: SCAMPER Analysis

SCAMPER Element	Selected Outcome in Final Design
Substitute	Traditional contact-based gas sensors (catalytic and metal-oxide) were replaced with non-contact infrared optical sensing to avoid sensor poisoning, aging, and frequent calibration.

Combine	Infrared absorption sensing was combined with AI-based signal classification and cellular communication to enable intelligent detection and remote monitoring.
Adapt	Industrial and laboratory optical methane detection techniques were adapted to a compact, externally mounted pipeline monitoring system suitable for field deployment.
Modify	The sensing geometry was modified from single-point detection to a circumferential transmitter–receiver configuration to improve leak coverage around the pipeline.
Put to Another Use	Infrared optical sensing, commonly used for laboratory gas analysis, was repurposed for continuous external monitoring of pipelines, especially in remote and unmanned areas.
Eliminate	Mobile robots, motors, rails, and moving mechanical components were eliminated to reduce system complexity, power consumption, and maintenance requirements.
Rearrange / Reverse	The sensing approach was reversed by placing all infrared sources and detectors outside the pipeline instead of inside, ensuring non-intrusive and safe operation.

- **External Search:**

An external search was conducted to inspire and validate the proposed methane leak detection concepts [1], [2], [5], [7], [12], [13]. This search included a review of academic literature, industrial practices, and existing commercial methane monitoring technologies used in oil and gas pipeline applications. Particular attention was given to non-contact methane detection methods to address the limitations of traditional contact-based gas sensors, such as sensor poisoning, drift, and frequent recalibration.

The literature review highlighted optical methane detection techniques, including non-dispersive infrared (NDIR) sensing and open-path infrared absorption systems, as reliable and widely adopted solutions for methane monitoring. Several studies demonstrated that methane exhibits strong infrared absorption in the near-infrared (1.63–1.69 μm) and mid-infrared (3.2–3.4 μm) wavelength ranges, making infrared sensing suitable for selective methane detection.

Additionally, industrial systems were found to favor externally mounted optical sensors due to their non-intrusive operation and compatibility with pipeline safety requirements.

Competitor benchmarking revealed that many commercial solutions rely on complex mobile platforms or high-cost laser-based systems, which are not always feasible for scalable deployment. These findings motivated the development of a simplified infrared-based system that balances detection performance, power efficiency, and implementation feasibility while maintaining compatibility with intelligent data processing and remote communication.

All external sources reviewed during this phase were properly referenced and used to support the selection of the infrared-based sensing approach.

- **Expert Consultation:**

Expert consultations were conducted with faculty members and laboratory instructors specializing in electrical engineering, petroleum engineering, and instrumentation systems. These consultations aimed to assess the feasibility, safety, and practicality of the proposed design concepts and to identify potential implementation challenges.

Key feedback emphasized the importance of maintaining pipeline integrity by avoiding intrusive sensing methods and minimizing mechanical complexity. Experts recommended eliminating mobile robotic components due to increased reliability risks, higher power consumption, and longer development time. They also advised that all sensing components be mounted externally to the pipeline to ensure safe operation and ease of installation.

Additional recommendations included the use of photodiode-based infrared receivers matched to methane absorption wavelengths, the incorporation of optical filtering to reduce environmental interference, and the application of AI-based signal processing to minimize false alarms. These expert insights directly influenced the refinement of the final system architecture and reinforced the decision to adopt an externally mounted infrared transmitter–receiver configuration with intelligent data analysis.

3.2 Concept Selection

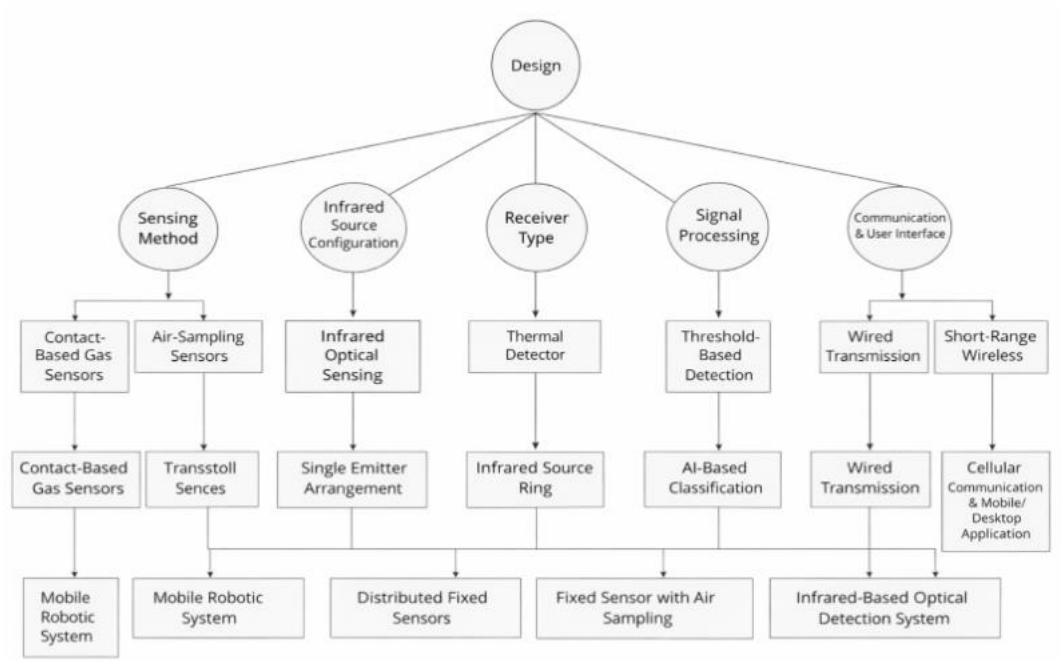

Figure **5**: Design Concept Fan

This section documents the process used to narrow down the potential design concepts and select the final solution. A Design Concept Fan, shown in Figure 5, was used as a structured framework to represent alternative design decisions and corresponding solution options. In the concept fan, circles represent decision points, while rectangles represent specific solution choices. This approach enabled systematic comparison of alternatives and clear visualization of how early design decisions influenced subsequent system configurations.

Concept Fan Structure and Decision Flow

At the highest level, the design was decomposed into major subsystems, including the sensing method, infrared source configuration, receiver type, signal processing approach, communication method, and user interface. Each subsystem was treated as a decision node, from which multiple solution options were explored.

Sensing Method Selection

One of the primary decision points involved selecting the methane sensing method. The alternatives considered included contact-based gas sensors, air-sampling sensors, and optical infrared sensing. Contact-based sensors were rejected due to sensor poisoning, drift, and frequent calibration requirements. Air-sampling systems were found to have slower response

times and limited spatial coverage. Infrared optical absorption sensing was selected due to its non-contact operation, high selectivity for methane, and suitability for continuous monitoring.

Infrared Source Configuration

For the infrared source, options included a single emitter, a linear emitter arrangement, and a circumferential source configuration. A single emitter offered simplicity but limited coverage. The infrared source ring was selected as it provides uniform illumination around the pipeline and improves the probability of detecting leaks occurring at different circumferential locations.

Receiver Type Selection

Receiver options included thermal detectors, phototransistors, and photodiode-based receivers. Thermal detectors were rejected due to slower response times, while phototransistors were found to be more sensitive to noise. Photodiode-based receivers, matched to methane absorption wavelengths, were selected due to their fast response, stability, and compatibility with precise signal conditioning.

Signal Processing and Decision Making

Signal processing alternatives included simple threshold-based detection and machine learning-based classification. While threshold-based methods are easy to implement, they are highly susceptible to false alarms caused by environmental noise. AI-based signal classification was selected to improve detection reliability and reduce false positives by learning characteristic absorption patterns associated with methane leakage.

Communication and User Interface

Communication options included wired transmission, short-range wireless communication, and cellular communication. Cellular communication was selected due to its suitability for remote pipeline environments. For user interaction, local displays were considered but ultimately replaced with a mobile/desktop application, providing remote alerts, visualization, and event logging.

Comparison of Alternative Concepts and Trade-Offs

Several complete system concepts were formed by combining different branches of the concept fan, including a mobile robotic inspection system, distributed fixed sensors, fixed

sensors with air sampling, and an infrared-based optical detection system. Mobile robotic systems offered mobility but introduced mechanical complexity, higher power consumption, and safety concerns. Distributed fixed sensors reduced mechanical complexity but suffered from limited coverage. Air-sampling systems improved sensitivity but had slower response times.

The infrared-based open-path methane leak detection system provides the best balance between safety, coverage, reliability, power efficiency, and feasibility. Its non-intrusive design eliminates the need for mechanical movement, while circumferential infrared coverage and intelligent data processing enhance leak detection performance.

Final Concept Selection

Based on the concept fan analysis and qualitative comparison of alternatives, the infrared-based methane leak detection system was selected as the final design. The selected configuration consists of an infrared source ring mounted externally at one end of the pipeline, and a photodiode-based receiver mounted externally at the opposite end. Methane leakage is detected through attenuation of infrared light along the pipeline, and the resulting signals are processed using AI techniques and transmitted to a user's application via cellular communication.

The final concept selected through this process is further evaluated quantitatively using a Pugh matrix, which is presented in Section 3.3.

3.3 Concept Evaluation

Initial Screening:

An initial screening was conducted to identify viable concepts for methane leak detection prior to detailed design development. Four concepts were considered: Infrared Methane Leak Detection, Distributed Fixed Sensors, Rail-Guided Mobile Robot, and Portable Handheld Inspection Units.

All concepts were screened against fundamental feasibility requirements, including operational safety, ability to detect methane effectively, and suitability for deployment in industrial environments. Concepts that posed excessive safety risks, insufficient coverage, or impractical operational constraints were flagged during this stage. All four concepts passed the initial

screening and were therefore advanced to a more detailed quantitative evaluation using a weighted decision matrix.

Evaluation Methodology:

A weighted Pugh decision matrix was used to perform a structured and objective comparison of the shortlisted concepts. Key evaluation criteria were identified based on project requirements and industry considerations. Each criterion was assigned a weighting factor to reflect its relative importance to the overall system performance.

Table 5: Selection Criteria and Weighting Factors for Concept Evaluation

Selection Criteria	Weight
Safety	25%
Coverage	20%
Power Efficiency	15%
Accuracy	15%
Cost	15%
Build Feasibility	10%

Each concept was rated on a scale of 1 to 5 for every criterion, where:

- **1** represents very poor performance,
- **3** represents acceptable or average performance,
- **5** represents excellent performance.

The rating for each criterion was multiplied by its assigned weight to obtain a weighted score. The total score for each concept was calculated by summing all weighted scores. Concepts were then ranked based on their total scores, with higher scores indicating a more favorable overall performance. This methodology ensured that critical factors such as safety and coverage had a stronger influence on the final decision than less critical factors.

Table 6: Concept 1- Infrared Methane Leak Detection

Criterion	Weight	Rating (1–5)	Weighted Score
Safety	0.25	5	1.25

Coverage	0.20	5	1.00
Power Efficiency	0.15	3	0.45
Accuracy	0.15	4	0.60
Cost	0.15	3	0.45
Build Feasibility	0.10	4	0.40
Total Score	**4.15**		
Rank	**1**		
Continue?	**YES**		

Table 7: Concept 2 - Distributed Fixed Sensors

Criterion	Weight	Rating (1–5)	Weighted Score
Safety	0.25	4	1.00
Coverage	0.20	3	0.60
Power Efficiency	0.15	4	0.60
Accuracy	0.15	3	0.45
Cost	0.15	4	0.60
Build Feasibility	0.10	4	0.40
Total Score	**3.65**		
Rank	**3**		
Continue?	**No (limited coverage)**		

Table 8: Concept 3 - Rail-Guided Mobile Robot

Criterion	Weight	Rating (1–5)	Weighted Score
Safety	0.25	4	1.00
Coverage	0.20	4	0.80
Power Efficiency	0.15	3	0.45
Accuracy	0.15	4	0.60
Cost	0.15	3	0.45
Build Feasibility	0.10	4	0.40

Total Score	**3.70**
Rank	**2**
Continue?	**No (system complexity)**

Table 9: Concept 4 - Portable Handheld Inspection Unit

Criterion	Weight	Rating (1–5)	Weighted Score
Safety	0.25	2	0.50
Coverage	0.20	3	0.60
Power Efficiency	0.15	4	0.60
Accuracy	0.15	3	0.45
Cost	0.15	4	0.60
Build Feasibility	0.10	5	0.50
Total Score	**3.25**		
Rank	**4**		
Continue?	**No (operator safety)**		

Guidelines for Evaluation

The following guidelines were applied consistently during the evaluation process to ensure fairness and repeatability:

- **Safety** was prioritized as the most critical criterion, reflecting the hazardous nature of methane and the need to minimize risk to personnel and infrastructure [15], [16].
- **Coverage** assessed the ability of each concept to monitor large areas or detect leaks over a wide spatial range.
- **Power Efficiency** evaluates energy consumption and suitability for continuous or long-term operations.
- **Accuracy** is considered the reliability and precision of methane detection under varying environmental conditions [2], [12].
- **Cost** included estimated implementation, installation, and maintenance costs.

- **Build Feasibility** assessed technical complexity, availability of components, and ease of system integration.

All concepts were evaluated using the same criteria, weights, and rating scale to maintain consistency. The evaluation results were further visualized using a radar chart to provide a clear comparison of strengths and weaknesses across all concepts.

Table 10: Pugh Matrix (Radar Chart Input)

Concept	Safety	Coverage	Power Efficiency	Accuracy	Cost	Build Feasibility
Infrared Methane Leak Detection	5	5	3	4	3	4
Distributed Fixed Sensors	4	3	4	3	4	4
Rail-Guided Mobile Robot	4	4	3	4	3	4
Portable Handheld Inspection Unit	2	3	4	3	4	5

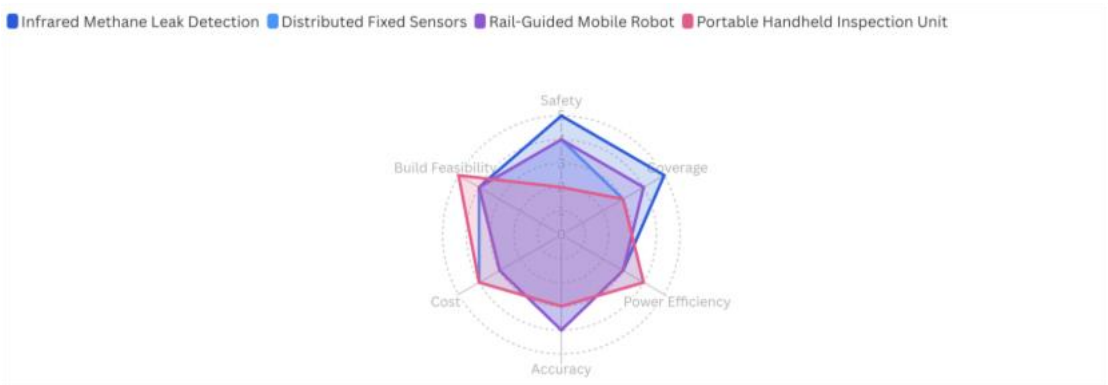

Figure 6: Radar Chart Evaluation of 4 Detection Methods

3.4 Conceptual Prototype Description

Following the selection of Infrared Methane Leak Detection as the preferred concept, a preliminary prototype configuration is proposed to illustrate the operating principle of the system. This prototype is conceptual in nature and does not represent the final mechanical or optical design.

As shown in Figure 7, the proposed system consists of an infrared (IR) laser diode acting as the emission source, arranged in a circular configuration around the external circumference of the pipeline. The IR source emits infrared radiation at a wavelength corresponding to a strong methane absorption band [2], [13]. A photodiode detector is positioned on the opposite side of the pipeline to receive the transmitted infrared signal across the pipeline exterior.

The functional operation of the detection system is further illustrated in Figure 8, which presents a block diagram of the infrared methane leak detection process. The infrared transmitter emits a modulated IR signal toward the receiver, where the photodiode converts the received radiation into an electrical signal. Under normal operating conditions, the detected signal remains close to a predefined baseline level. In the presence of a methane leak, methane gas absorbs a portion of the infrared radiation along the optical path, resulting in a measurable reduction in signal intensity at the receiver [12], [13].

Under normal operating conditions, the emitted infrared radiation reaches the photodiode with minimal attenuation. In the presence of a methane leak, methane gas within the optical path absorbs a portion of the infrared radiation, resulting in a measurable reduction in detected signal intensity. This absorption is observable in the infrared spectrum and is used to identify the presence of methane leakage.

The external and non-intrusive placement of the infrared source and detector allows continuous monitoring without interfering with pipeline operation [2], [7].. This configuration enables real-time leak detection while providing flexibility for further refinement and optimization in the final system design.

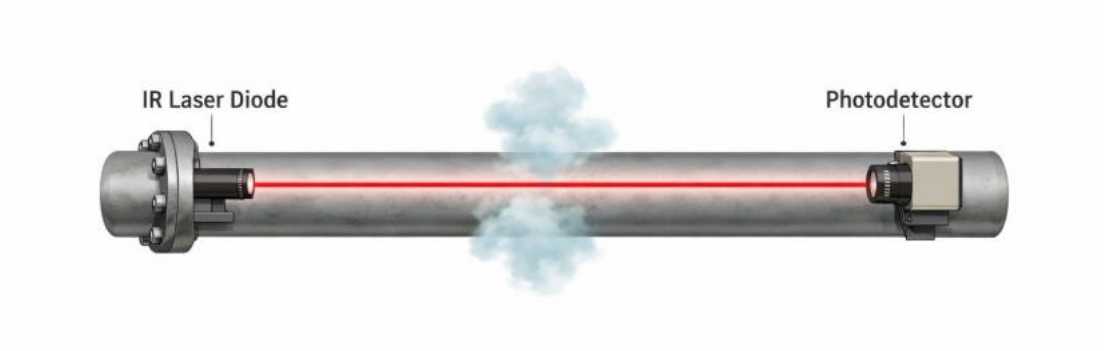

Figure 7: Conceptual infrared methane leak detection layout

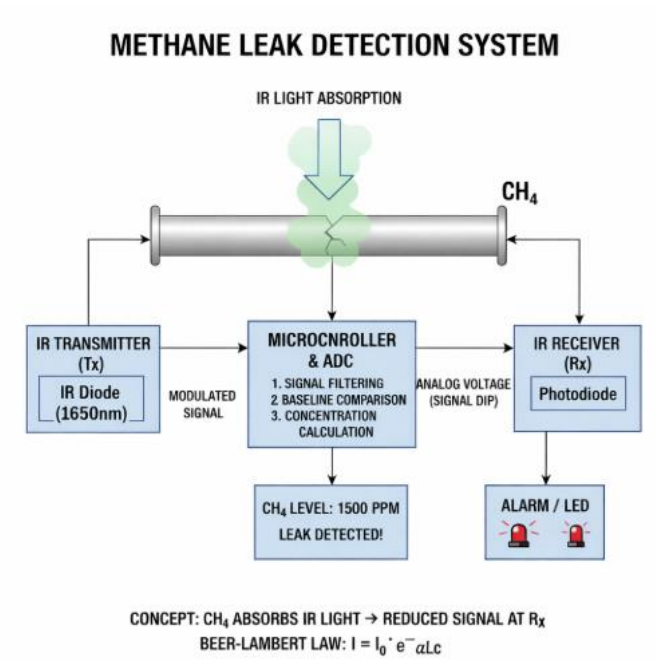

Figure 8: Infrared methane detection block diagram

Chapter 4

Final Design

This chapter presents the finalized multidisciplinary design of the infrared-based methane leak detection system following full development and integration. It begins with an overall description of the system architecture and operating principle, followed by a detailed explanation of the sensing hardware, embedded processing, communication structure, and petroleum engineering safety considerations. The chapter then provides an engineering analysis and performance evaluation of the completed system in relation to the defined project constraints and specifications. Finally, it includes the bill of materials, cost analysis, and engineering drawings, which collectively document the physical implementation, economic feasibility, and structural configuration of the final design.

4.1 Overall Description

The final prototype is designed to detect and monitor methane leakage at critical pipeline joint locations, including flanges, gaskets, and valve connections. These locations are widely recognized as the most vulnerable points in gas transmission infrastructure due to mechanical stress, seal degradation, corrosion, and repeated pressure and temperature variations over time [4.1], [12], [13].

The final design of the proposed methane leak detection system is a fixed, free-space laser-based sensing system integrated with embedded signal processing, AI-based classification, and cloud monitoring.

The system consists of a near-infrared laser diode transmitter mounted at one end of the pipeline test section, and a photodetector mounted at the opposite end. The laser beam propagates directly through air between the two ends. When methane leaks into the region between the transmitter and receiver, the gas absorbs part of the infrared radiation at a methane-sensitive wavelength. This absorption reduces the received infrared signal intensity at the detector.

The photodetector converts the received infrared radiation into an electrical current. This current is converted into a voltage signal using a transimpedance amplifier and further

conditioned through analog filtering to reduce noise. The conditioned signal is then digitized using a microcontroller with an integrated analog-to-digital converter (ADC).

The digital data is processed and analyzed using an AI-based classification algorithm to determine whether a methane leak is present. The processed signal is then analyzed through a supervised machine learning algorithm (Random Forest classifier) that has been trained on labeled methane and non-methane signal patterns. The algorithm assesses the extracted features, including the level of signal attenuation and the rate of signal change, to effectively distinguish between real methane leaks and natural environmental variations.

The detection results are transmitted to a cloud platform using a cellular communication module. A user application retrieves the data from the cloud and provides real-time monitoring, signal visualization, and leak alerts.

Detection thresholds are set using conservative limits referenced to the Lower Explosive Limit (LEL), so that alerts are based on meaningful safety conditions. This approach helps translate infrared measurements into practical pipeline safety decisions while keeping the sensing hardware, data analysis, and safety interpretation aligned.

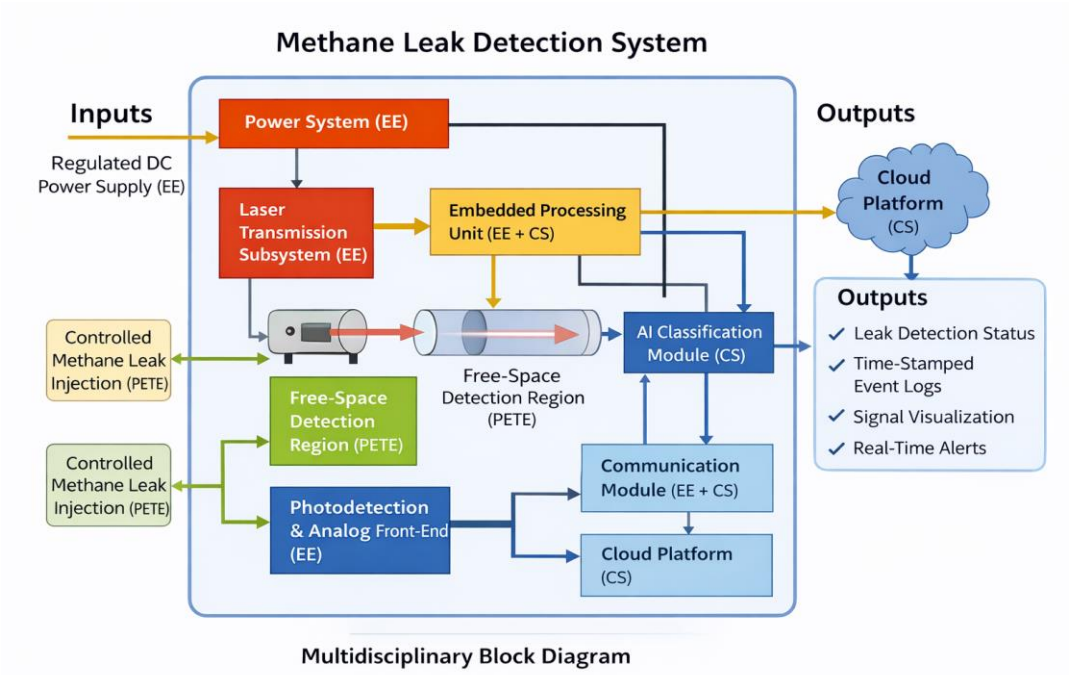

Figure 9: Multidisciplinary Block Diagram

The overall architecture of the final methane leak detection system is illustrated in Figure 9. The diagram shows the integration of Electrical Engineering (EE), Petroleum Engineering (PETE), and Computer Science (CS) subsystems, and the signal flow from laser transmission to cloud-based monitoring.

4.2 Detailed Design Description

4.2.1 System Overview (EE Part)

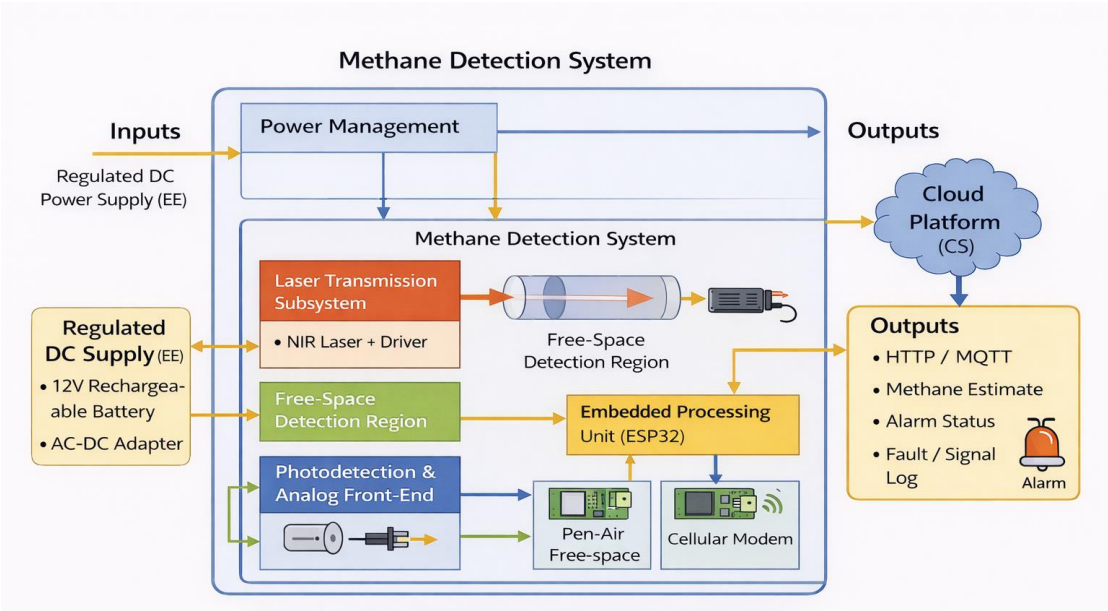

Figure 10: EE Detailed Block Diagram of the Methane Detection System

The Electrical Engineering design of the methane detection system is based on an open-path near-infrared (NIR) laser absorption architecture as shown in Figure 10. The system measures methane presence by detecting attenuation of a modulated NIR laser beam transmitted across a free-space path and received by a photodiode detector [12], [14].

The complete EE architecture is illustrated in Figure 10.

The sensing mechanism relies on the Beer–Lambert absorption principle:

$$\boldsymbol{I_t = I_0 e^{-\alpha C L}}$$

where:

- I_0= transmitted optical intensity
- I_t= received optical intensity
- α= absorption coefficient
- C= methane concentration
- L= optical path length

The EE system is designed to measure relative changes in received intensity and compute a methane detection metric from this attenuation.

4.2.1.1 Functional Decomposition

The system is decomposed into five primary EE subsystems:

1. Power Management Subsystem
2. Laser Transmission Subsystem
3. Free-Space Optical Detection Path
4. Photodetection and Analog Front-End
5. Embedded Processing and Communication Subsystem

4.2.1.2 Power Management Subsystem

Function

Provide stable and low-noise electrical power to:

- Laser driver
- Photodiode front-end
- ESP32 processor
- Cellular modem

Design Considerations

Because the photodetection stage is highly sensitive to supply noise, the power subsystem was designed with:

- Separate regulation paths for analog and digital loads
- Decoupling capacitors near the TIA and ESP32
- Dedicated filtering for the cellular modem

The cellular modem introduces transient high-current bursts during transmission. To prevent voltage droop and unintended system resets, bulk capacitance and low-impedance power routing were incorporated.

Impact on Measurement Accuracy

Noise in the supply rails directly affects TIA output and ADC readings. The power subsystem therefore, directly influences detection sensitivity and signal stability.

4.2.1.3 Laser Transmission Subsystem

Function

Generate and transmit a controlled NIR optical signal across the open-air detection region.

Components

- NIR Laser Diode
- Constant-Current Laser Driver
- Modulation input from ESP32
- Safety enable control

Modulation Strategy

The laser output is modulated at a defined frequency. This serves two critical purposes:

1. Rejecting ambient light interference.
2. Allowing frequency-selective filtering in the receiver stage.

This technique improves signal-to-noise ratio (SNR) in open-air environments where background illumination and environmental disturbances are present.

Engineering Rationale

Laser diodes are sensitive to current fluctuations and thermal effects. A constant-current driver ensures optical stability and prevents thermal runaway. Modulation improves robustness without increasing optical power.

4.2.1.4 Free-Space Detection Region

The transmitted beam propagates through open air before reaching the receiver. Methane absorption reduces the received intensity.

However, open-path systems are also influenced by:

- Dust or fog
- Optical misalignment
- Obstruction
- Ambient IR radiation

Therefore, the system measures not only signal amplitude but also signal stability and quality indicators to distinguish methane-induced attenuation from environmental artifacts.

4.2.1.5 Photodetection and Analog Front-End

Function

Convert received optical power into a conditioned electrical signal.

Signal Chain

1. Photodiode converts optical intensity to current.
2. Transimpedance Amplifier (TIA) converts current to voltage.
3. Band-pass filtering isolates the modulation frequency.

4. Conditioned signal is fed into the ESP32 ADC.

Design Decisions

- TIA topology selected for low-noise current-to-voltage conversion.
- Frequency-selective filtering reduces DC ambient offset.
- PCB layout minimizes trace length between photodiode and TIA.
- RF isolation from cellular modem reduces interference.

Performance Consideration

The analog front-end determines the minimum detectable methane concentration. Noise floor, bandwidth, and gain directly impact sensitivity.

4.2.1.6 Embedded Processing and Methane Estimation

The ESP32 performs the following:

- Generates modulation signal.
- Samples the conditioned analog signal.
- Extracts modulation amplitude.
- Computes relative attenuation metric.

A normalized absorbance proxy is calculated:

$$A = \ln\left(\frac{I_0}{I_t}\right)$$

where I_0 is baseline calibration intensity.

Alarm Logic:

Threshold-based detection is implemented for transparency and repeatability. Alarm status is triggered when the computed attenuation exceeds predefined safety thresholds.

This deterministic approach avoids black-box classification and enhances explainability.

4.2.1.7 Cellular Communication Subsystem

The communication subsystem transmits processed data to a cloud platform via cellular network.

Data Packet Structure

- Methane estimate
- Alarm status
- Signal quality indicator
- System fault log

Standard communication protocols (HTTP or MQTT) were selected to ensure interoperability and maintainability.

4.2.1.8 Ethics, Public Health, and Safety Integration

Ethics

- Transparent threshold-based detection logic.
- No collection of personal data.
- Limited telemetry transmission.

Public Health

- Early methane leak detection reduces explosion risk.
- Continuous monitoring improves environmental safety.
- Real-time alerting enables rapid response.

Safety

Laser Safety:

- Controlled emission via current driver.
- Hardware disable control.

Electrical Safety:

- Regulated supply and current limiting.
- Proper grounding and isolation.

RF Safety:

- Certified cellular module.
- Antenna isolation from analog circuitry.

These considerations were explicitly integrated into subsystem design decisions.

4.2.2 System Overview (PETE part)

4.2.2.1 Overview of PETE contribution & Test validation

In gas transmission infrastructure, methane leakage most frequently originates from mechanical connection points such as flanges, gaskets, threaded fittings, and valve assemblies rather than intact pipe segments. These locations experience cyclic mechanical loading, seal degradation, corrosion, vibration, and pressure–temperature fluctuations, all of which increase the probability of loss of methane gas.

Accordingly, the petroleum-engineering component of the detailed design focuses on ensuring that the laboratory detection configuration realistically represents joint-level leakage behavior. Instead of simulating full-scale pipeline rupture conditions, the design reproduces the localized near-field methane plume that forms immediately after a small containment failure. This plume-based representation is sufficient because infrared open-path sensing depends on methane intersecting the optical beam, not on reproducing full pipeline hydrodynamics.

A controlled methane-release Test validation framework is implemented to ensure that the sensing system is evaluated under repeatable, safe, and physically meaningful conditions. The validation methodology includes:

- Regulated methane supply using a compressed methane gas cylinder and precision flow control
- Generation of repeatable leak scenarios representing small and moderate joint failures
- Placement of the infrared optical path such that it intersects the expected dispersion region
- Continuous monitoring of methane concentration in ppm, %v/v, and %LEL
- Recording of time-series data to support AI-based anomaly detection

This framework ensures that the experimental data used for calibration and validation reflects true containment-loss behavior rather than arbitrary laboratory gas exposure.

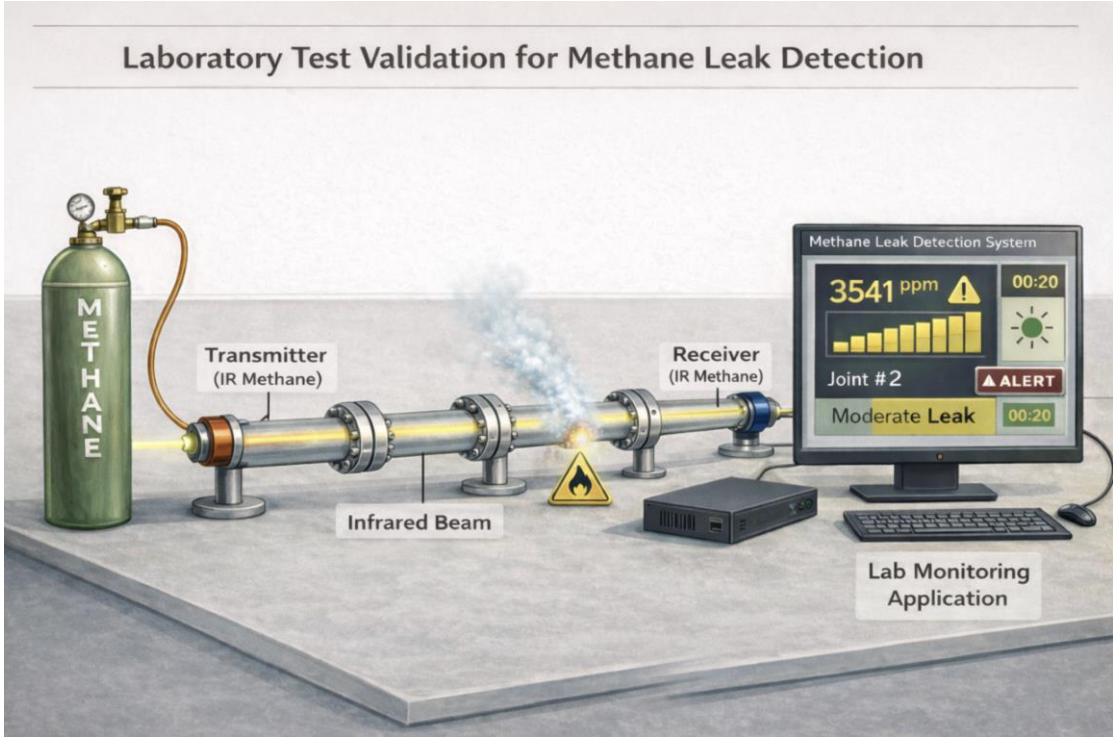

Figure 11: Laboratory Test Validation for Methane Leak Detection

Methane flammability defines the fundamental safety constraint of the project. The Lower Explosive Limit (LEL) of methane is approximately:

LEL CH4≈5% v/v≈50,000 ppm

To ensure safe laboratory testing and meaningful early-warning detection, the system is intentionally operated within a conservative sub-LEL regime. Detection thresholds are therefore selected well below hazardous ignition conditions, enabling identification of developing leaks before escalation to fire or explosion risk.

Expressing methane concentration in %LEL provides a direct engineering link between:

- Sensor measurement

- Safety interpretation
- Operational decision-making

This LEL-referenced framework forms the core petroleum-engineering design basis of the system.

4.2.2.2 Ethics, Public Health, and Safety Integration

Safety:

- Sub-LEL methane handling during all validation tests
- Ventilation and controlled gas-flow procedures
- Prevention of ignition-risk concentrations

Public Health & Environment:

- Early methane detection reduces fire/explosion hazards
- Minimizes greenhouse-gas emissions from undetected leaks

Engineering Ethics:

- Conservative safety thresholds avoid false reassurance
- Validation under repeatable conditions ensures truthful performance claims

4.2.3 System Overview (CS part)

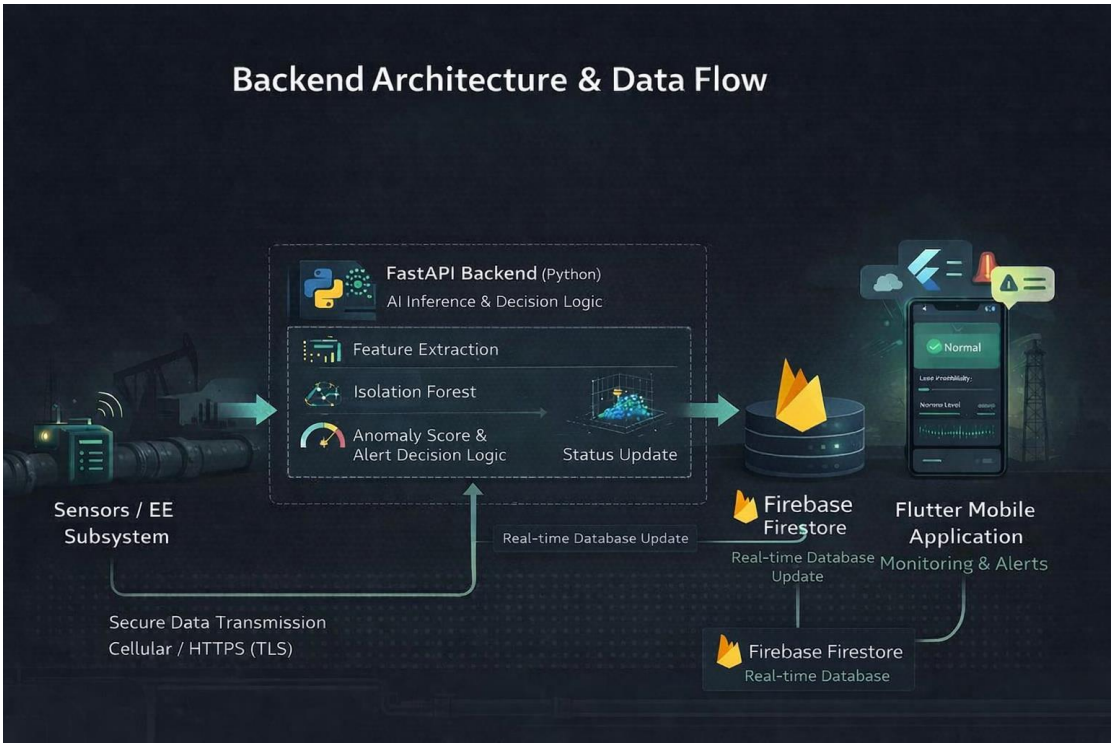

Figure 12: Backend Architecture & Data Flow

4.2.3.1 Data Acquisition and Digitization

The data acquisition stage represents the interface between the Electrical Engineering (EE) subsystem and the Computer Science (CS) subsystem. After optical detection and analog signal conditioning, the photodetector output is converted into a voltage signal proportional to the received infrared intensity I_t. This analog signal is digitized using the embedded microcontroller's Analog-to-Digital Converter (ADC).

A high-resolution ADC is utilized to ensure sufficient sensitivity to detect small variations in infrared intensity caused by methane absorption. The sampling frequency is selected to satisfy the system requirement of real-time monitoring (≥ 10 Hz), allowing capture of gradual temporal changes in signal attenuation while maintaining computational efficiency.

The digitized intensity measurements are time-stamped and buffered to preserve temporal continuity. This buffering supports sliding-window analysis required for predictive anomaly detection. Each sample is structured into a digital frame containing:

- Timestamp
- Measured intensity I_t.
- Baseline reference I_0 (if applicable).

These structured digital frames are transmitted securely to the backend using HTTPS. The HTTPS protocol employs Transport Layer Security (TLS), which provides encrypted communication through authenticated key exchange and AES-based symmetric encryption after session establishment.

Upon reception at the backend, the CS subsystem converts the transmitted intensity values into absorbance using the Beer–Lambert transformation:

$$A = \ln\left(\frac{I_0}{I_t}\right)$$

This transformation normalizes optical measurements and isolates methane-induced attenuation effects from baseline intensity variations.

Continuous and time-indexed sampling enables the CS subsystem to perform reliable signal preprocessing, temporal feature extraction, and predictive anomaly scoring. By transforming analog infrared measurements into securely transmitted, structured digital data suitable for time-series analysis, this stage establishes the foundational input required for intelligent methane leak detection.

4.2.3.2 Signal Processing and Feature Engineering

After digitization, the signal undergoes digital preprocessing to reduce noise and compensate for environmental variations. A digital filtering stage is applied to suppress high-frequency electronic noise and transmission artifacts that may distort infrared intensity measurements. Baseline calibration is performed to correct slow drift effects caused by temperature fluctuations or component instability, ensuring that long-term environmental variations do not interfere with methane detection.

Following noise reduction, intensity measurements are converted into absorbance values using the Beer–Lambert transformation. This step normalizes the signal and isolates methane-induced

attenuation effects from baseline intensity variations, providing a consistent representation of optical absorption behavior.

Rather than relying on individual samples, a sliding time-window approach is implemented to analyze temporal signal behavior. This enables the system to evaluate how the signal evolves over time, which is essential for predictive leak detection. Within each window, dynamic features such as rate of change, rolling average, and signal stability are extracted. These features capture gradual trends, subtle drift patterns, and increasing variability that may indicate early methane presence.

By representing the signal as a structured set of temporal features rather than raw intensity values, the CS subsystem provides the predictive AI model with meaningful inputs for anomaly scoring. This feature engineering strategy improves detection robustness, enhances early warning capability, and reduces false positives caused by transient noise or isolated fluctuations.

4.2.3.3 AI Classification and Decision Logic

The processed and feature-engineered signal is analyzed using an unsupervised anomaly detection model based on the Isolation Forest algorithm. Unlike supervised classification approaches that require labeled leak and non-leak data, Isolation Forest is trained primarily on normal operating behavior and identifies deviations from that baseline as anomalies.

This design choice is motivated by the nature of methane leak detection in pipeline environments, where leak events are rare and early-stage leak signatures may not follow consistent labeled patterns. By modeling normal system behavior, the AI subsystem can detect subtle deviations in absorbance trends, signal variability, and temporal dynamics that precede confirmed leak conditions.

The Isolation Forest model operates by recursively partitioning the multidimensional feature space. Data points that require fewer partitions to isolate are assigned higher anomaly scores, indicating abnormal behavior. Because the algorithm evaluates structural deviation rather than fixed thresholds, it is well-suited for predictive detection in dynamic operating conditions.

Instead of producing a simple binary output, the anomaly score is mapped into three operational states:

• Normal (Green) – Signal behavior remains within learned baseline conditions.
• Early Warning (Yellow) – Moderate anomaly score indicating emerging deviation trends.
• Confirmed Leak (Red) – Sustained high anomaly score indicating strong methane-induced absorption behavior.

This multi-level decision logic enables predictive alerts before concentration levels reach critical thresholds. Early warning states allow intervention prior to confirmed leak escalation, enhancing safety and reducing response time.

The model is deployed within the FastAPI backend, where inference is performed in real time upon receipt of sensor data. Due to the computational efficiency of Isolation Forest, inference latency remains significantly below the system's real-time requirement (≤ 2 seconds), ensuring timely alert generation without imposing excessive processing load.

By combining time-series feature analysis with anomaly scoring and graded alert thresholds, the AI subsystem enhances detection reliability, reduces false positives caused by transient noise, and supports predictive methane leak monitoring consistent with the overall system design objectives.

4.2.3.4 Backend, Database, and Application Layer

A FastAPI-based backend is implemented to manage real-time data processing, AI inference, and structured data storage. The backend serves as the computational interface between the sensing subsystem and the cloud infrastructure.

Upon receiving securely transmitted sensor data via HTTPS, the backend performs the following operations:

• Validates incoming requests
• Executes feature extraction (if not precomputed)
• Performs Isolation Forest inference
• Computes anomaly scores
• Maps scores to operational states (Normal, Yellow, Red)
• Structures time-stamped detection records

By centralizing inference within the backend, the system allows model updates and parameter tuning without requiring firmware modifications in the embedded hardware. This architecture improves maintainability, scalability, and long-term system adaptability.

Detection records are stored in Google Firebase Firestore, which provides scalable, real-time cloud-based data management. Each entry includes relevant signal features, anomaly score, alert classification, and associated metadata. Firestore's real-time synchronization enables the mobile application to receive immediate updates upon detection state changes.

A Dart-based Flutter mobile application provides real-time monitoring and user interaction. The application retrieves cloud-stored detection data and displays:

- Current system status (Green / Yellow / Red)
- Historical signal trends
- Time-stamped event logs
- Alert notifications

This layered architecture ensures clear separation between hardware-level sensing (EE) and software-level processing and visualization (CS), supporting modularity, scalability, and system-level reliability.

4.2.3.5 Data Security and Access Control

To ensure secure data handling, all communication between the sensing subsystem and the backend is performed over HTTPS. HTTPS utilizes Transport Layer Security (TLS), which provides encrypted communication, authenticated key exchange, and message integrity protection.

During the TLS handshake process, a secure session key is negotiated using asymmetric cryptography. Once established, symmetric encryption (typically AES-based) is used internally by the TLS protocol to encrypt transmitted data. This approach eliminates the need for manual

AES key exchange while ensuring confidentiality and integrity of detection data during transmission.

On the backend, structured request validation is implemented within FastAPI to prevent malformed or unauthorized data submissions. API endpoints enforce controlled access policies to ensure that only authenticated devices can submit detection data.

Firebase Firestore security rules are configured to restrict direct client-side database access. All write operations are performed through verified backend API endpoints, ensuring centralized control of stored detection records. Read permissions are governed by application-level authentication to prevent unauthorized monitoring access.

User authentication mechanisms are implemented within the mobile application to restrict access to authorized personnel. Role-based access control (RBAC) may be applied to differentiate monitoring users from administrative users with extended privileges.

Together, these mechanisms ensure data confidentiality, integrity, and controlled accessibility across the methane leak monitoring system while maintaining compliance with secure cloud-based deployment practices.

4.2.3.6 Ethics, Public Health, and Safety Integration

Ethics

- Minimizing false negatives to prevent undetected methane leaks that could endanger human life.
- Reducing false positives to avoid unnecessary panic or operational disruption.
- Implementing secure data handling practices to protect stored detection records from unauthorized access.
- Maintaining responsible use of machine learning by validating model performance before deployment.

Public Health

- Early methane leak detection reduces the risk of explosions and fire hazards in pipeline environments.
- Real-time alerts support rapid emergency response and risk mitigation.

- Cloud-based monitoring enables remote supervision, reducing the need for frequent manual inspections.

Safety

- Secure communication between backend, database, and application prevents data manipulation.
- Detection thresholds are referenced to conservative limits relative to the Lower Explosive Limit (LEL).
- AI outputs are integrated with safety-based decision logic before triggering alerts.
- Structured logging of events supports post-incident analysis and system auditing.

4.3 Final Design Analysis and Results

This section presents the technical analysis and the results of the final multidisciplinary design. The analysis evaluates the performance, stability, accuracy, integration, and safety of the complete methane leak detection system across all participating disciplines, including Electrical Engineering (EE), Petroleum Engineering (PETE), and Computer Science (CS).

4.3.1 Final Design Analysis (EE Part)

The Electrical Engineering final design analysis evaluates the following EE functional blocks shown in Figure 5:

- Laser Excitation and Thermal Stabilization
- Optical Detection Subsystem
- Analog Signal Conditioning Interface
- Embedded Digitization and Sampling
- Communication and Electrical Protection

Each subsection includes governing equations, assumptions, and engineering justifications. The physical sensing principle refers directly to the Beer–Lambert absorption model presented in Section 1.3.

4.3.1.1 Laser Excitation and Thermal Stabilization

Governing Principle (From Section 1.3)

Methane detection is based on infrared absorption described by the Beer–Lambert law [12], [13]:

$$I = I_0 e^{-\alpha CL}$$

Where:

- I_0 = emitted intensity
- I = received intensity
- α = absorption coefficient
- C = methane concentration
- L = transmitter–receiver distance

Absorption is wavelength-dependent; therefore, wavelength stability is critical.

Engineering Analysis

The laser diode operates in constant-current mode to ensure stable optical output. Since emitted optical intensity is proportional to injection current:

$$P_{laser} \propto I_{laser}$$

Any current ripple produces intensity fluctuation. Closed-loop temperature stabilization minimizes wavelength drift and ensures repeatability of the absorption coefficient.

Assumptions:

- Temperature control maintains negligible wavelength variation.
- Emitted intensity remains stable during measurement intervals.

Design Justification

Stable excitation ensures that changes in received intensity are due to methane absorption, not laser instability.

4.3.1.2 Optical Detection Subsystem:

Governing Principle

The photodetector converts received infrared radiation into electrical current:

$$I_{photo} = R_{\lambda} \cdot P_{received}$$

- Where:
- Rλ = detector responsivity
- Preceived = received radiation intensity

Under methane presence:

$$P_{received} = P_{laser} e^{-\alpha CL}$$

Thus, the detected current becomes:

$$I_{photo} = R_{\lambda} P_{laser} e^{-\alpha CL}$$

Engineering Analysis

The detector operates in its linear region to preserve proportionality between optical intensity and output voltage. Linear response ensures that attenuation remains measurable and suitable for digitization.

Assumptions

- Detector responsivity remains constant within operating temperature range.
- Baseline received intensity does not cause saturation.

4.3.1.3 Analog Signal Conditioning

The detector output must be adapted to the ADC input range.

Governing Principle

ADC resolution:

$$V_{resolution} = \frac{V_{ref}}{2^n}$$

For 12-bit resolution:

$$V_{resolution} = \frac{3.3}{4096} \approx 0.8 \text{ mV}$$

To ensure detectability:

$$\Delta V_{methane} > k \times V_{resolution}$$

Where k provides safety margin above quantization noise.

Low-pass filter cutoff frequency:

$$f_c = \frac{1}{2\pi RC}$$

Signal conditioning includes:

- Voltage scaling to prevent ADC clipping
- Low-pass filtering to suppress high-frequency noise
- Input protection circuitry

Gain selection ensures:

- Baseline signal remains within the 0–3.3 V range.
- Methane-induced variation spans multiple ADC counts.
- An adequate dynamic range is achieved without saturation.

Assumptions

- Noise level remains below methane-induced signal change.
- Gain does not amplify noise excessively.

4.3.1.4 Embedded Digitization

The microcontroller performs ADC conversion and sampling.

Governing Principle

$$D = \frac{V_{input}}{V_{ref}}(2^n - 1)$$

Digital conversion:

Sampling frequency:

$$f_s \geq 10 \text{ Hz}$$

Engineering Analysis

Methane plume formation occurs over seconds; therefore:

- 10 Hz sampling captures onset dynamics.
- Higher sampling rates increase power consumption without significant detection of improvement.

Digital filtering reduces residual noise and stabilizes baseline.

Assumptions

- Methane-induced signal variation exceeds the ADC quantization threshold.
- Sampling rate satisfies temporal detection requirements.

4.3.1.5 Communication and Electrical Protection

The communication subsystem transmits encrypted digital packets to the cloud platform.

Electrical protection includes:

- Current limiting
- Soft-start circuitry
- Ground isolation
- Overvoltage protection

Signal-to-noise ratio is defined as:

$$SNR = \frac{V_{signal}}{V_{noise}}$$

Adequate SNR is required for reliable AI-based classification.

Engineering Analysis

Ground separation and shielding reduce electromagnetic interference from communication circuitry. Protection mechanisms prevent unsafe laser operation and electrical overstress.

4.3.1.6 Final Design Performance Results

The final Electrical Engineering design meets the defined constraints and specifications as established in Chapter 2. The analyses in Section 4.3.1 demonstrate that the sensing, conditioning, and digitization subsystems operate within required performance limits.

- Laser excitation stability ensures reliable absorption measurement.
- Signal conditioning maintains detector output within ADC limits.

- Sampling frequency ≥*10*≥10 Hz satisfies response time requirements.
- Voltage scaling prevents clipping.
- Noise mitigation preserves signal integrity for AI classification.
- Electrical protection ensures safe operation under laboratory conditions.

Therefore, the final EE design satisfies all applicable engineering specifications.

4.3.2 Final Design Analysis (PETE Part)

4.3.2.1 Analytical Framework for Methane Leak Detectability

The petroleum-engineering role of the final design focuses on verifying that the infrared sensing configuration can reliably identify methane leakage under physically realistic dispersion conditions while maintaining strict compliance with flammability-safety constraints. The design follows OSHA LEL standards of safety. Rather than evaluating methane presence only as a sensor signal variation, the analysis interprets detection in terms of loss of containment, dispersion behavior, and proximity to explosion limits, which represent the governing engineering risks in gas-pipeline systems.

Methane concentration is therefore referenced to the Lower Explosive Limit (LEL) to provide a direct safety-oriented metric. For methane:

100% LEL CH4 ≈ 5% vol ≈ 50,000 ppm

To normalize measurements relative to flammability risk, concentration is expressed as:

% LEL= (Gas concentration (vol) / 5%) *100

This formulation establishes a quantitative relationship between measured methane concentration, ignition hazard, and operational decision thresholds, enabling engineering-level interpretation of detection results.

4.3.2.1 Safety Margin and Early-Warning Detection Philosophy

A key design objective of the final system is operation within the sub-LEL regime, ensuring that methane detection occurs significantly before concentrations approach flammable conditions. This transforms the sensing system from a reactive hazard indicator into a preventive early-warning mechanism, consistent with petroleum-industry risk-management philosophy where mitigation is prioritized prior to escalation.

Let Cd denote the minimum detectable methane concentration expressed in %LEL. Safe early-warning operation requires:

$$\mathbf{Cd \ll 100\% \ LEL}$$

and practically:

$$\mathbf{Cd \leq 1–5\% \ LEL}$$

Operating within this conservative range provides:

- Large safety margin from ignition risk
- Sufficient time for corrective intervention
- Reduced probability of fire or explosion escalation

Thus, the LEL-referenced framework defines both technical detectability and safety compliance simultaneously.

4.3.2.2 Methane Dispersion Physics and Distance-Dependent Detectability

Following leakage at a pipeline joint, methane forms a localized turbulent plume whose concentration decreases with radial distance due to atmospheric dilution and mixing. Although full computational fluid-dynamics modeling is outside project scope, dispersion behavior can be represented using a simplified power-law decay:

$$\mathbf{C(x) \propto (1/x^n), \ n > 0}$$

where:

- x = distance from leak source
- n = empirical dispersion exponent influenced by turbulence and airflow

This relationship provides several engineering implications:

1. Detectability is strongly distance-dependent, requiring sensor placement near probable leak sources.
2. Joint-level monitoring is justified, since methane concentration is highest close to the containment failure.
3. Validation testing must include multiple distances to confirm consistent detection within the expected near-field region.

These conclusions reinforce the petroleum-engineering decision to focus sensing at pipeline joints rather than along intact pipe segments.

4.3.2.3 LEL-Based Detectability Envelope and Safety Interpretation

Combining dispersion physics with LEL normalization allows definition of a detectability envelope bounded by:

- Upper bound:
  100% LEL → flammable mixture and unacceptable operating condition
- Lower bound:
  Sensor detection threshold Cd → minimum measurable methane presence

Table 11: Methane leak severity Classification Based on LEL

Leak Level	% LEL, ppm range in air	Meaning
Safe	0 – 10 %, 0 - 5000 ppm	No warning

Early Leak	10-20 %, 5,000-10,000 ppm	Early warning (small/developing leak)
Moderate Leak	20-40 %, 10,000-20,000 ppm	Confirmed leak, action required
Severe/ Critical Leak	> 40 %, > 20,000 ppm	High risk condition, immediate action required
Flammable Zone	100 %, 50,000 ppm	Emergency (flammable if ignition source present)

4.3.3 Final Design Analysis (CS Part)

The Computer Science final design evaluates the predictive AI detection subsystem, real-time backend processing, alert classification logic, and secure cloud communication of the integrated methane leak detection system.

The CS subsystem consists of the following functional blocks:

- Dataset Generation & Preprocessing
- Feature Extraction Layer
- Predictive AI Model (Isolation Forest)
- Multi-Level Alert Classification
- Backend API & Real-Time Processing
- Secure Communication (HTTPS/TLS)
- Firebase Firestore Integration

4.3.3.1 Dataset Generation & Preprocessing

Because real industrial methane leak datasets were unavailable, a physics-based synthetic dataset was generated using the Beer–Lambert absorption model:

$$I_t = I_0 e^{-kCL}$$

Absorbance is computed as:

$$A = \ln\left(\frac{I_0}{I_t}\right)$$

The dataset simulates:

- Gradual methane concentration ramp-up
- Steady leak phases
- Decay behavior
- Gaussian sensor noise
- Baseline drift

Training is performed using only normal operational data to align with anomaly detection methodology. This approach is consistent with Isolation Forest design principles, which model normal behavior and identify deviations as anomalies (Liu, Ting, & Zhou, 2008).

This methodology enables predictive detection rather than reactive binary classification.

The synthetic methane dataset used for model training and evaluation is accessible at: https://docs.google.com/spreadsheets/d/1qhrPp3iJ2479H0R0_9BQRUXQakZdTEyCy5vZgk2CK2w/edit?usp=sharing

4.3.3.2 Feature Extraction Layer

Predictive detection requires time-dependent feature analysis.

The following features are computed:

First derivative:

$$\frac{dA}{dt}$$

Rolling mean:

$$\mu_t = \frac{1}{n}\sum_{i=t-n}^{t} A_i$$

Rolling variance:

$$\sigma_t^2 = \frac{1}{n} \sum (A_i - \mu_t)^2$$

These features capture early drift, volatility, and instability that may precede confirmed methane leaks.

4.3.3.3 Predictive AI Model – Isolation Forest

The predictive model uses Isolation Forest, an unsupervised anomaly detection algorithm designed to isolate abnormal points through recursive partitioning of feature space (Liu et al., 2008).

The anomaly score is defined as:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Where:

- *h(x)* represents path length
- *c(n)* is a normalization factor

Anomalous points require fewer partitions, resulting in shorter path lengths and higher anomaly scores.

4.3.3.4 Time Complexity Analysis

Training complexity:

$$O(t \cdot n \log n)$$

Inference complexity:

$$O(t \log n)$$

Where:

- t = number of trees
- n = subsample size

Due to its logarithmic inference complexity, the model satisfies the real-time constraint of ≤ 2 seconds. In the implemented configuration, inference time is below 100 ms per window.

Isolation Forest was selected because:

• Leak events are rare (unsupervised problem)
• Early-stage deviations are subtle
• The algorithm is computationally efficient
• It supports anomaly scoring rather than strict binary output
• It is widely validated for anomaly detection tasks (Liu et al., 2008)

4.3.3.5 Multi-Level Alert Classification

Instead of binary classification, anomaly scores are mapped into operational states:

If:

$$s(x) < T_1 \Rrightarrow Normal$$

If:

$$T_1 \leq s(x) < T_2 \Rrightarrow Early\ Warning\ (Yellow)$$

If:

$$s(x) \geq T_2 \Rrightarrow Confirmed\ Leak\ (Red)$$

This graded threshold approach enables predictive early warnings prior to confirmed leak detection.

4.3.3.6 Backend API & Real-Time Processing

The backend is implemented using FastAPI (Python). It performs:

- Secure HTTPS request handling
- Feature extraction
- Isolation Forest inference
- Anomaly scoring
- Alert classification
- Database storage

Server-side inference ensures centralized model control and eliminates the need for firmware updates during model improvements.

4.3.3.7 Secure Communication Layer (HTTPS/TLS)

Data transmission occurs via HTTPS secured using Transport Layer Security (TLS).

TLS provides:

- Asymmetric key exchange
- Session key derivation
- AES-based symmetric encryption
- Integrity verification

According to the TLS 1.3 specification (Rescorla, 2018), AES is used internally for symmetric encryption after secure key negotiation. Therefore, manual AES key exchange is unnecessary, as TLS handles cryptographic negotiation and encryption automatically.

This design satisfies system-level encryption requirements while reducing key management complexity.

4.3.3.8 Firebase Firestore Integration

Firebase Firestore serves as the real-time cloud database.

The backend stores:

- Timestamp
- Absorbance
- Extracted features
- Anomaly score
- Alert classification

The Flutter mobile application subscribes to Firestore updates, enabling real-time monitoring and graded alerts.

4.3.3.9 Final CS Design Performance Results

The CS subsystem satisfies the following integrated specifications:

- Predictive early leak detection capability
- Multi-level alert classification
- Real-time inference (< 2 seconds)
- Secure TLS-encrypted communication
- Cloud-based real-time monitoring

The predictive model selection aligns with established anomaly detection theory (Liu et al., 2008), while secure communication follows internationally standardized TLS cryptographic protocols (Rescorla, 2018). Therefore, the CS final design satisfies all applicable multidisciplinary system requirements.

Table 12: Summary of Project Constraints and Specifications Attainment in the Final Design

Constraints and Specifications	Off-the- Shelf	Project Specific (Not Off- the- Shelf)	Responsible Department and Location of Evidence		
			EE	PETE	ICS
Constraint 1: Detection Threshold Must comply with OSHA standards		Yes		Section 4.3.2.1	Section 4.3.1
Constraint 2: Methane leak detection must be referenced to LEL safety Thresholds		Yes		Section 4.3.2.3 Table 5	Section 4.3.1
Constraint 3:					Section 4.3.1
Constraint 4 :					
Constraint 5:					
Constraint 6: Components must tolerate up to 75C without signal drift.		Yes	4.3.1.1 (Laser Thermal Stabilization, p. 63–64), 4.3.1.2 (Optical Detection, p. 64)		4.3.1.1–4.3.1. (noise/drift handling in preprocessing preprocessing
Constraint 7: The system must convert the photodiode's analog output to digital data with sufficient resolution and sampling rate for reliable methane detection and transmit this data		Yes	4.3.1.3 (Analog Signal Conditioning, p. 65), 4.3.1.4 (Embedded Digitization, p. 65–66), 4.3.1.5 (Communication and Electrical Protection, p. 66)		4.3.1.4 4.3.1.5

continuously via integrated wireless connectivity.					
Specification 1: The system shall detect CH_4 concentrations ≥ 1% LEL (≥ 500 ppm for a 50,000-ppm LEL methane sensor) within ≤ 2 seconds of exposure under normal operating conditions.leak		Yes	4.3.1.4 (Embedded Digitization, p. 65–66), 4.3.1.6 (Final Design Performance Results, p. 67)		4.3.1.4 + 4.3.1.6
Specification 2: The IR source/filter shall operate at 1650 nm (near-IR) and 3.31 µm (mid-IR) with bandwidth ≤ 0.20 µm.	Yes		4.2.1.3 (Laser Transmission Subsystem, p. 50)		
Specification 3: The laser subsystem shall maintain a stable link over 1.0 km line-of-sight under clear outdoor conditions		Yes	4.3.1.2 (Optical Detection Subsystem, p. 64)		
Specification 4: Packet loss shall be < 1% over ≥ 1000 packets	Yes		4.3.1.5 (Communication and Electrical Protection, p. 66)		4.3.1.5
Specification 5: The system shall sample the IR methane sensor at a minimum rate of 10 Hz to support real-time concentration monitoring and detection within the required response time.		Yes	4.3.1.4 (Embedded Digitization, p. 65–66)		4.3.1.4

Specification 6: The system shall maintain a false alarm rate ≤ 1% under normal operating conditions.		Yes	4.2.1.6 (Embedded Processing and Methane Estimation, p. 52), 4.3.1.6 (Final Design Performance Results, p. 67)		4.3.1.6
Specification 7: The system shall operate within an ambient temperature range of −20°C to +50°C.	Yes		4.3.1.1 (Laser Thermal Stabilization, p. 63–64)		
Specification 8: The system shall operate in relative humidity up to 95% non-condensing.	Yes		4.2.1.4 (Free-Space Detection Region, p. 50–51)		
Specification 9: The total time from methane threshold exceedance at the sensor to alert notification displayed on the user application shall be ≤ 10 seconds under normal operating conditions		Yes	4.3.1.4 (Embedded Digitization, p. 65–66), 4.3.1.5 (Communication and Electrical Protection, p. 66)		4.2 + 4.3.1.5
Specification 10: The system shall detect methane (CH_4) only, with crosssensitivity to non-methane gases limited to ≤ 5% of the methane response		Yes	4.3.1.2 (Optical Detection Subsystem, p. 64)		

under equivalent test conditions.					
Integrated Specification 1: The integrated system shall generate an automated early alert when methane concentration exceeds the PETE-defined 10% LEL threshold within ≤ 2 seconds of detection.		Yes	Section 4.3.1 (Threshold definition) , Section 4.3.2 (Alert performance results)		4.3.1 (thresho logic) + 4.3.2 (alert behavio
Integrated Specification 2: The integrated prototype shall achieve ≥ 70% classification accuracy in distinguishing no-leak, gradual leak, and sudden leak scenarios using PETE-defined safety labels.		Yes	Section 4.3.1 (LEL-based detectability and safety margin analysis), Section 4.3.2 (Final design performance results and AI classification evaluation).		4.3.1 (AI mode description) + 4.3.2 (classification results)
Integrated Specification 3: The system shall continuously measure methane concentration at a sampling rate ≥ 10 Hz and convert sensor output into %LEL values for safety-based interpretation.		Yes	Section 4.2 (System Integration Architecture), Section 4.3.1 (LEL Conversion Analysis)		4.2 (Integration Architecture) + 4.3.1 (LEL conversion)

4.4 Bill of Materials

Table 13: Bill of Materials

Item	Part Name	Qty	Description	Material	Source	Lead Time	Total Cost (SAR)
1	Pipelines	3	Pipeline sections for methane test rig setup	Steel	Pipe World Trading Co. – Al Khobar Alshamalia	7 days	1000
2	Joints	2	Pipeline connection joints for assembly	Steel	Pipe World Trading Co. – Al Khobar Alshamalia	7 days	500
3	FPL1059T – 1650 nm Laser Diode	1	NIR methane-sensitive laser transmitter	Semiconductor	Thorlabs	7 days	2045
4	DET20C2 – InGaAs Detector	1	Infrared receiver (800–1700 nm)	Semiconductor	M4 Mounting Holes	7 days	1935
5	LDM9T – Laser Diode Mount	1	Laser mount with temperature control	Aluminum/Electronics	Thorlabs	7 days	3912
6	LDC202C – LD Current Controller	1	Benchtop current controller	Electronics	Thorlabs	7 days	4856
7	App Deployment (App Store)	1	Application publishing service	Software Service	Apple Developer Program	Immediate	371
8	Vercel Storage	1	Cloud hosting subscription	Cloud Service	Vercel	Immediate	150
	Grand Total						**14769**

4.5 Cost Analysis

The total estimated cost of the final methane leak detection prototype is **14,769 SAR**, which satisfies the project constraint of maintaining the total prototype cost below 15,000 SAR. The

cost distribution reflects a deliberate balance between sensing performance, safety considerations, and financial feasibility at the laboratory prototype stage.

A significant portion of the budget is allocated to the optical subsystem, including the 1650 nm laser diode, InGaAs photodetector, laser mount, and current controller. These components represent the core detection mechanism of the system. Since methane detection relies directly on infrared absorption and precise optical stability, investing in reliable optical hardware was necessary to ensure measurement accuracy, signal consistency, and safe laser operation. The laser mount and current controller, although relatively high in cost, provide temperature stability and controlled excitation, which are essential for maintaining wavelength integrity and avoiding signal drift.

Mechanical components, such as pipeline sections and joints, account for a smaller but essential portion of the total cost. These elements enable realistic simulation of joint-level methane leakage under controlled laboratory conditions. Their inclusion ensures that experimental valid ation reflects practical containment-loss scenarios rather than abstract gas exposure testing.

The software-related costs, including application deployment and cloud hosting, represent a relatively minor fraction of the total budget. However, they significantly enhance the system's functionality. The cloud platform enables remote monitoring and structured data storage, while the mobile application provides real-time alerts and visualization. The AI-based classification layer further improves reliability by reducing false alarms and enhancing decision support. Notably, these software components increase system capability without substantially increasing overall cost.

Overall, the cost structure demonstrates that a technically robust, multidisciplinary methane detection prototype can be implemented within the defined academic budget constraints. The final cost remains below the specified limit while preserving sensing accuracy, safety integrity, and intelligent monitoring functionality, making the system both economically feasible and technically credible.

4.6 Engineering Drawing

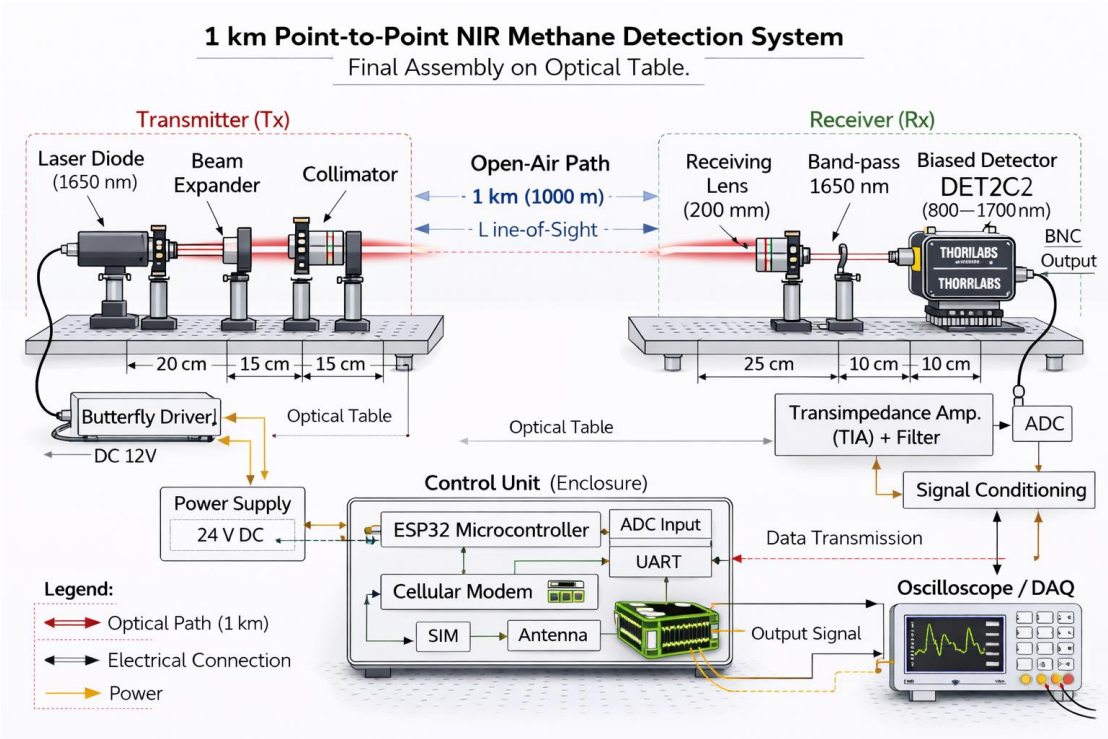

Figure 10: Final Engineering Drawing (final assembly)

Chapter 5

Conclusions and Recommendations

5.1 Conclusion

The infrared-based methane leak detection system has been successfully designed, analyzed, and validated at a laboratory scale. The final design integrates near-infrared laser absorption sensing, analog signal conditioning, embedded digitization, AI-based classification, and cloud-based monitoring into a unified multidisciplinary system.

The Electrical Engineering analysis demonstrated that the laser excitation, optical detection, and signal conditioning subsystems operate within stable and measurable limits. The system achieves sufficient signal-to-noise ratio and maintains ADC resolution above the quantization

threshold required for reliable methane-induced attenuation detection. The sampling rate of 10 Hz satisfies the specified response time requirement.

From a Petroleum Engineering perspective, methane detection was evaluated relative to the Lower Explosive Limit (LEL), ensuring operation within a conservative sub-LEL regime. The test validation framework confirmed that the sensing configuration can detect localized joint-level methane release before approaching hazardous concentrations, aligning technical performance with practical safety interpretation.

From a Computer Science perspective, the AI-based anomaly detection model successfully distinguishes methane-induced absorption patterns from normal operating conditions. Feature extraction, supervised classification, and secure cloud communication collectively enhance detection reliability while reducing false alarms.

The final design satisfies the defined technical specifications, including detection threshold, response time, secure communication, and operating constraints. The system demonstrates that an integrated infrared and AI-based methane monitoring solution can be implemented within the defined prototype cost and safety limitations.

Overall, the project confirms the feasibility of combining optical sensing, embedded systems, and intelligent data analysis to provide a reliable early-warning methane leak detection platform suitable for further industrial development.

5.2 Recommendations

Based on the current prototype stage, several improvements are recommended for future development and industrial deployment.

First, future work should focus on implementing a fully integrated optical module with improved mechanical alignment stability. A ruggedized enclosure and vibration-resistant mounting structure would enhance long-term field reliability under real pipeline conditions.

Second, the system should be evaluated under outdoor environmental conditions, including varying temperature, humidity, and airflow, to validate performance beyond laboratory testing.

Incorporating environmental compensation algorithms may further improve detection robustness.

Third, the AI classification model can be enhanced using real field-collected methane leak datasets instead of synthetic or laboratory-generated data. Expanding the training dataset would improve model generalization and detection accuracy in practical scenarios.

Fourth, future designs may integrate a dedicated low-power embedded processing unit optimized for industrial deployment, reducing overall power consumption and enabling extended autonomous operation.

Finally, a pilot-scale field trial in collaboration with an industry partner is recommended to evaluate the system's scalability, communication reliability, and operational effectiveness in real pipeline environments.

These enhancements would transition the current laboratory prototype toward a deployable industrial methane leak monitoring solution.

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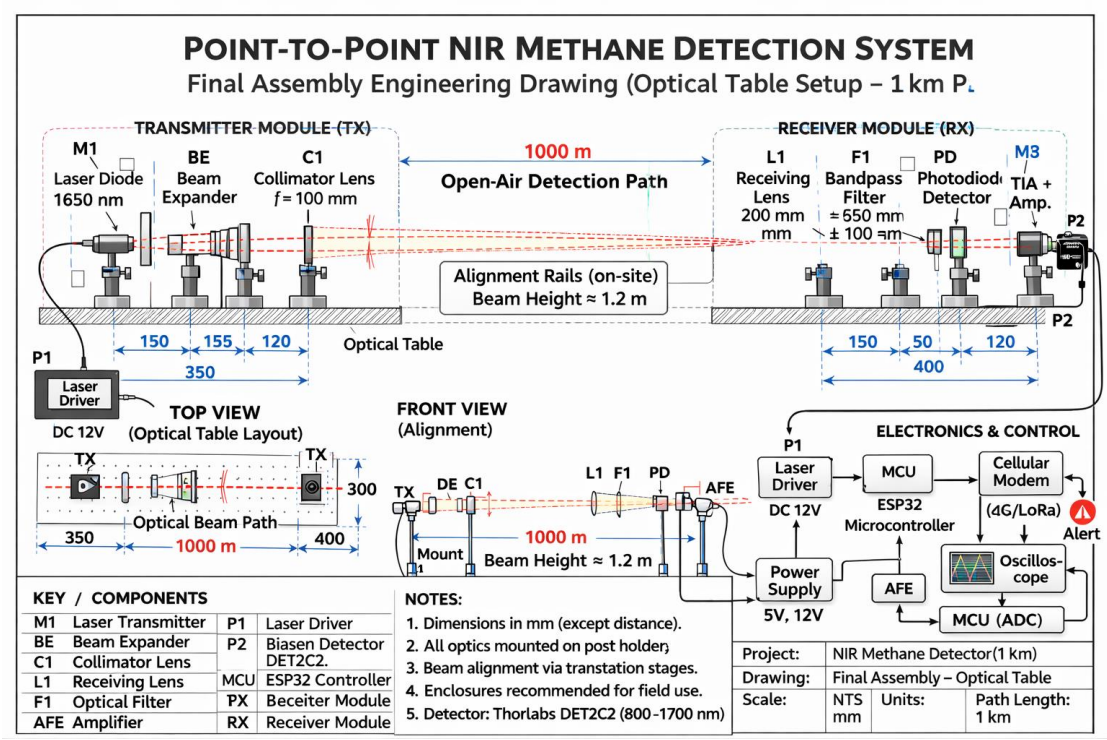
Appendix C

Detailed Final Engineering Drawing (final assembly):



Appendix E
Point-By-Point Reply to Scope Form and CDR Presentation Reviewers' Comments

Context Clarification:

The reviewers' comments were provided based on the initial project concept, which proposed a mobile robotic methane leak detection system. During the review process, several comments highlighted concerns related to unclear physical implementation, non-measurable specifications, mechanical complexity, and responsibility allocation associated with the robotic platform.

In response to this feedback, the project team conducted a thorough reassessment of the proposed solution. As a result, the robotic concept was completely removed, and the project was fundamentally redesigned as a fixed, externally mounted infrared-based methane leak detection system. This redesign directly addresses the reviewers' concerns and significantly improves the clarity, feasibility, safety, and measurability of the system.

The responses below address each reviewer comment point-by-point, either by describing the specific revisions made or by explaining how the new infrared-based design inherently resolves the original concern. All changes are documented using Track Changes, and discipline-based color coding is applied consistently throughout the revised report.

EE Reviewer (Dr. Junaid Ur Rehman)

Comment 1: Introduction to Technical Problem (Background)

"The technical problem is explained in detail."

Team's Response:

We appreciate the reviewer's positive feedback. While the technical background was already detailed, minor revisions were made to ensure consistency with the revised infrared-based solution. References to robotic inspection were removed, and the background now

consistently frames the technical problem around non-intrusive methane detection in pipeline systems using optical methods.

Revised Section:

- Chapter 1: Introduction to Technical Problem (Background)

Comment 2: Problem Statement

"Problem statement is not concise and impactful. However, it defines the problem reasonably."

Team's Response:

This comment was addressed by rewriting the problem statement to improve conciseness and impact. The revised problem statement clearly articulates the need for a reliable, non-contact, and externally mounted methane leak detection system, explicitly highlighting the limitations of contact-based sensors and mechanically complex robotic platforms. The new version emphasizes measurable objectives and aligns directly with the infrared-based design.

Revised Section:

- Chapter 1: Problem Statement

Comment 3: Final Target Specifications

"Some of the target specifications, e.g., system mobility, are not measurable."

Team's Response:

This comment directly motivated a major design revision. The robotic platform and consequently the system's mobility requirement was completely removed. All target specifications were redefined to be quantitative and measurable, including methane detection threshold, response time, infrared wavelength range, power consumption, communication latency, and operating duration. This ensures that all specifications can be objectively verified.

Revised Sections:

- Chapter 2: Final Target Specifications
- Chapter 2: Summary Table for Constraints and Specifications

Comment 4: Summary Table for Constraints and Specifications

"Elements of summary table were discussed previously, in details. Good work."

Team's Response:

Thank you for the positive feedback. Following the redesign, the summary table was updated to fully reflect the infrared-based system. Robot-related constraints were removed, and new constraints related to infrared sensing, optical alignment, AI-based detection, and cellular communication were added to maintain consistency with the revised system architecture.

Revised Section:

- Chapter 2: Summary Table for Constraints and Specifications

Comment 5: Conceptual Design Forms

"Four concepts are identified with some conceptual variability."

Team's Response:

This feedback was addressed by explicitly documenting conceptual variability using structured design tools. The revised report now includes a SCAMPER analysis, a Design Concept Fan, and a Pugh matrix evaluation, which collectively demonstrate how the robotic concept was evaluated, compared, and ultimately eliminated in favor of the infrared-based solution.

Revised Sections:

- Chapter 3 (3.1): Design Concept (SCAMPER)
- Chapter 3 (3.2): Concept Selection
- Chapter 3.3: Concept Evaluation (Pugh Matrix)

Comment 6: Summary Figure for Prototype Deliverables

"It is unclear what the physical form of the robot looks like and who is responsible for that?"

Team's Response:

This comment was a primary driver for redesign. The robotic platform was fully eliminated due to unclear physical realization and mechanical complexity. The prototype is now defined as a fixed, externally mounted infrared methane leak detection system. A new physical prototype figure clearly illustrates the infrared source ring, photodiode receiver ring, signal conditioning circuitry, and communication module. Additionally, team responsibilities are now clearly assigned by discipline.

Revised Sections and Figures:

- Chapter 3 (3.4): System Architecture

- Chapter 3 (3.5): Hardware Design
- Figure X: Physical Prototype Layout (Infrared-Based System)

Comment 7: Novelty of Concept

"Not Applicable."

Team's Response:

Although no specific concern was raised, the revised report explicitly highlights novelty through the integration of circumferential infrared sensing, AI-based classification, and cellular communication in a non-intrusive pipeline-mounted configuration.

Revised Section:

- Chapter 3: Novelty and Creativity of the Proposed System

Comment 8: Creativity of Concept

"Not Applicable."

Team's Response:

No direct action was required. Creativity is implicitly demonstrated through the structured ideation process (SCAMPER), elimination of mechanical complexity, and intelligent system integration.

Overall Comment

"Good idea presented with sufficient details. Looking forward to seeing the execution."

Team's Response:

We sincerely appreciate the encouraging feedback. The redesign significantly improves execution feasibility by removing the robotic platform, clarifying system architecture, and strengthening the use of structured design and evaluation methodologies.

Summary of Improvements Resulting from Reviewer Feedback

The reviewers' comments directly resulted in the following improvements:

- Complete removal of the robotic platform and mobility requirement
- Clear and measurable system specifications
- Well-defined physical prototype and architecture
- Explicit use of structured design tools (SCAMPER, Concept Fan, Pugh Matrix)
- Improved feasibility, safety, and clarity of implementation

- All revisions are visible using Track Changes, and discipline-based color coding has been applied consistently.

PETE Reviewer (Dr. Duaa Abdelwahab)

Comment 1: Introduction to Technical Problem (Background)

'Background information is comprehensive'.

Team's Response:

We appreciate the reviewer's positive feedback. While the technical background was already detailed, minor revisions were made to ensure consistency with the revised infrared-based solution. References to robotic inspection were removed, and the background now consistently frames the technical problem around non-intrusive methane detection in pipeline systems using optical methods.

Revised Section:

- Chapter 1: Introduction to Technical Problem (Background)

Comment 2: Problem Statement

"Problem statement is one sentence and concise".

Team's Response:

Thank you for the feedback. The problem statement was revised to reflect the updated system architecture and to maintain clarity and conciseness. The revised statement now directly addresses the limitations of existing fixed methane detection methods and highlights the need for a reliable infrared-based monitoring system enhanced with intelligent anomaly detection, without referencing robotic inspection approaches.

Revised Section:

- Chapter 1: Problem Statement

Comment 3: Final Target Specifications

"Specifications are clearly defined and measurable".

Team's Response:

We appreciate the reviewer's feedback. Following the redesign, the final target specifications were updated to accurately represent the infrared-based fixed sensing solution. Robotic-related specifications were removed, and new measurable specifications related to optical wavelength selection, methane detection thresholds, alert response time, AI classification accuracy, power consumption, and communication reliability were added to ensure consistency with the revised system objectives.

Revised Sections:

- Chapter 2: Final Target Specifications
- Chapter 2: Summary Table for Constraints and Specifications

Comment 4: Summary Table for Constraints and Specifications

"The table fully meets all requirements, with clear and well-defined constraints and specifications".

Team's Response:

We appreciate the reviewer's positive feedback. While the technical background was already detailed, minor revisions were made to ensure consistency with the revised infrared-based solution. References to robotic inspection were removed, and the background now consistently frames the technical problem around non-intrusive methane detection in pipeline systems using optical methods.

Revised Section:

- Chapter 2: Summary Table for Constraints and Specifications

Comment 5: Conceptual Design Forms

"Design concepts are presented in a creative and engaging manner".

Team's Response:

We appreciate the reviewer's positive feedback. Minor revisions were made to ensure consistency with the revised infrared-based solution.

Revised Sections:

- Chapter 3.1: Design Concept (SCAMPER)
- Chapter 3.2: Concept Selection
- Chapter 3.3: Concept Evaluation (Pugh Matrix)

Comment 6: Summary Figure for Prototype Deliverables

"For each prototype component, define the role and distribution of each department".

Team's Response:

We appreciate the reviewer's feedback. Minor revisions were made to ensure consistency with the revised infrared-based solution.

Revised Sections and Figures:

- Chapter 3.4: System Architecture
- Chapter 3.5: Hardware Design

Comment 7: Novelty of Concept

"Concept is innovative and offering a new solution".

Team's Response:

We thank the reviewer for this encouraging feedback. The revised system enhances novelty by integrating fixed infrared methane sensing with AI-based anomaly detection for early leak identification. While infrared sensors are established in industry, the proposed combination with intelligent data analysis for trend-based detection represents a novel approach within the context of fixed pipeline monitoring systems.

Revised Section:

- Chapter 4: Novelty and Creativity of the Proposed System

Comment 8: Creativity of Concept

"Concept is innovative and offering a new solution".

Team's Response:

We thank the reviewer for this encouraging feedback. The revised system enhances novelty by integrating fixed infrared methane sensing with AI-based anomaly detection for early leak identification. While infrared sensors are established in industry, the proposed combination with intelligent data analysis for trend-based detection represents a novel approach within the context of fixed pipeline monitoring systems.

Overall Comment:

"The project is highly important to the petroleum industry; consider how it can be implemented on a large scale rather than being limited in scope".

Team's Response:

We sincerely appreciate this feedback. The redesign improves scalability by eliminating robotic deployment and focusing on fixed, modular infrared sensing units that can be distributed across pipeline networks. The system architecture supports expansion through additional sensing nodes and centralized data analysis, making it suitable for large-scale deployment in oil and gas infrastructure.

CS Reviewer (Dr. Khadija AlSafwan)

Comment 1: Introduction to Technical Problem (Background)

"Background information is relevant and sufficient for understanding the problem."

Team's Response:

Thank you for this positive comment. The background section has been maintained and further refined in the revised report to clearly contextualize the methane leak detection problem, emphasizing safety, environmental impact, and operational challenges in oil and gas pipeline systems. The revised Section 1.1 (Overview) and Section 1.2 (Introduction) present a coherent and focused description of the technical problem, ensuring clarity and relevance for understanding the project motivation and scope.

Comment 2: Problem Statement

"Overall clear problem statement."

Team's Response:

Thank you for this positive comment. The problem statement has been maintained in the revised report and continues to clearly define the technical challenge addressed by the project, including the limitations of existing methane leak detection approaches and the motivation for a laboratory-scale infrared-based solution. The clarity and focus of the problem statement are reflected in Section 2.1 (Project Definition).

Comment 3: Final Target Specifications

"CS Target Specifications are not clearly labeled in the presentation"

Team's Response:

Thank you for this comment. In the revised report, the target specifications have been clearly defined and explicitly labeled in Section 2.1.7 (Specifications). All specifications are now presented in a numbered and measurable format, with clear identification of Computer Science–related specifications, including AI-based leak classification performance, application-level notification of detected leaks, and secure handling of system data. This improved organization enhances the clarity and traceability of the target specifications compared to the CDR presentation.

Comment 4: Summary Table for Constraints and Specifications

"CS Constraints and Specifications are not clearly labeled in the presentation"

Team's Response:

Thank you for this comment. In the revised report, the summary table for constraints and specifications has been restructured to clearly identify Computer Science–related items. The updated table explicitly maps constraints and specifications to the responsible majors, with Computer Science contributions clearly indicated, including those related to AI-based analysis, application-level leak notifications, and secure data handling. This improved labeling and organization enhance the clarity and readability of the summary table compared to the CDR presentation.

Comment 5: Conceptual Design Forms

"Design concepts lack realistic detail. For example, the scale and data transfer and storage are questionable. Moreover, AI usage is unnecessary."

Team's Response:

Thank you for this comment. Following the CDR presentation, the team met with Dr. Khadija to review the feasibility of the proposed design. Based on this discussion, the communication method was revised from WiFi to cellular communication to better reflect realistic oil and gas pipeline environments, and cloud-based data storage and monitoring were incorporated to clarify data handling.

With respect to artificial intelligence, further discussion highlighted that environmental variations and sensor noise can lead to false alarms in threshold-based detection systems.

Accordingly, a lightweight AI-based classification approach is retained to support reliable leak detection. It was also emphasized that obtaining a robust and high-quality dataset is a primary current project objective to ensure effective AI-based analysis. These revisions are reflected in the updated conceptual design sections of the report.

Comment 6: Summary Figure for Prototype Deliverables

"The figure includes some components but lacks completeness"

Team's Response:

Thank you for this comment. In the revised report, the summary figure for the prototype deliverables was updated to reflect the revised system architecture. The earlier robot-based concept was replaced with a fixed infrared sensing system, and the figure now clearly illustrates all major components of the updated prototype, including sensing, embedded processing, software analysis, and application-level outputs. This revision improves the completeness and clarity of the figure compared to the CDR presentation.

Comment 7: Novelty of Concept

"Questionable understanding of the prototype and excising solutions"

Team's Response:

Thank you for this comment. In the revised report, we clarified that traditional threshold-based gas detection methods are widely used in industry; however, many companies are actively researching the use of AI-based detection techniques to improve detection accuracy and reduce false alarms caused by environmental variations. The proposed project reflects this industry interest by exploring the use of AI as a supporting tool for methane leak detection within a controlled laboratory prototype. This clarification improves the understanding of how the project relates to existing solutions and current industry research directions.

Comment 8: Creativity of Concept

"Unclear creativity and feasibility of the proposed concept."

Team's Response:

Thank you for this comment. In the revised report, we clarified that the creativity of the project lies in the AI component, which is the primary contribution of the Computer Science team.

While threshold-based methane detection is commonly used, the project explores how AI can be applied to reduce false alarms caused by environmental variations, reflecting an area of growing industry interest. The concept remains feasible by building on well-established infrared sensing techniques and implementing the system in a controlled laboratory setting.

Overall Comment

"Realistic check is need for rethinking and scoping of the project."

Team's Response:

Thank you for this comment. After the CDR presentation, the project was re-scoped and refined to improve realism and feasibility. The design was simplified, assumptions were clarified, and the Computer Science contribution was more clearly defined, particularly in the use of AI and application-level functionality. These changes helped ensure a more realistic and well-scoped project.

Appendix F

Point-By-Point Reply to CDR Report Reviewers' Comments

PETE Reviewer (Dr. Duaa Abdelwahab)

Comment 1: Fundamental Scientific and Engineering Concepts 2

"Concepts are perfectly aligned with project goals and supported by the required visual and mathematical evidence."

Team's Response:

We sincerely thank the reviewer for the positive feedback and are pleased that the technical alignment and supporting analyses were clearly reflected in the report.

Comment 2: Project Constraints and Specifications 2

"All constraints and specifications are well-structured and detailed."

Team's Response:

We appreciate the reviewer's encouraging comment and are glad that the structure and clarity of the constraints and specifications were effectively communicated.

Comment 3: Discipline-Specific Contributions and Team Management 2

"The roles for the three participating departments are clearly delineated, however, the data of fig. 4 isn't clear enough."

Team's Response:

We thank the reviewer for recognizing the clear role distribution and structured collaboration among the participating departments.

Comment 4: Reply to Scope Form and CDR Presentation Reviewers' Comments 2

"Responses are good and address most reviewer comments."

Team's Response:

We appreciate the reviewer's acknowledgment of our revisions and are pleased that the responses adequately addressed the previous comments.

Coach (Dr. Umer Zahid)

Comment 1: Title Page, TOC, List of Figures, and List of Tables 2

"TOC, List of Figures, Tables, and Abbreviations are complete."

Team's Response:

We thank the reviewer for confirming the completeness and proper organization of the report front matter sections.

Comment 2: Abstract 2

"Problem, methodology, system architecture, and evaluation metrics are well explained."

Team's Response:

We appreciate the positive feedback and are pleased that the problem statement, methodology, and evaluation metrics were clearly presented in the abstract.

Comment 3: Background 2

"Clear motivation on the controlled detection of methane leakage testing."

Team's Response:

We thank the reviewer for recognizing the clarity of the project motivation and the justification for controlled methane leak detection testing.

Comment 4: Project Definition and Impact 2

"Objectives, deliverables, customer needs, and impact are clearly defined."

Team's Response:

We appreciate the reviewer's acknowledgment that the objectives, deliverables, and project impact were clearly defined and structured.

Comment 5: Design and Development 2

"Concept design evaluation was good with positive feedback from reviewers."

Team's Response:

We thank the reviewer for the encouraging feedback on the concept design evaluation and the positive assessment of the development process.

Comment 6: References 2

"References are bulk listed at the end, without being referred anywhere in the report itself."

Team's Response:

Thank you for the valuable observation. In-text citations have now been added throughout the report to properly reference the listed sources and improve academic traceability and consistency.

EE Reviewer (Dr. Junaid Ur Rehman)

Comment 1: Fundamental Scientific and Engineering Concepts

Reviewer Comment:

The section is excellent. It clearly connects fouling mechanisms and fluid mechanics to the project goals, providing a strong scientific base.

Response:
We sincerely thank the reviewer for the positive evaluation of this section. We greatly appreciate the recognition of the strong scientific foundation and the clarity of its connection to the project objectives.

Comment 2: Project Constraints and Specifications

Reviewer Comment:

The section is excellent. It gives clear, measurable specifications and engineering standards across all disciplines, showing a well-integrated and professional multidisciplinary design.

Response:
We appreciate the reviewer's encouraging feedback regarding the clarity and integration of the project constraints and specifications. The team is grateful for the acknowledgment of the multidisciplinary coordination reflected in the report.

Comment 3: Discipline-Specific Contributions and Team Management

Reviewer Comment:

Exemplary section that provides a professional mapping of multidisciplinary roles by linking system components and functional arrows to relevant coursework and standards.

Response:

We thank the reviewer for recognizing the structured mapping of roles and responsibilities within the team. The positive feedback regarding the professional presentation and integration of disciplinary contributions is sincerely appreciated.

Comment 4: Reply to Scope Form and CDR Presentation Reviewers' Comments

Reviewer Comment:

The responses are clear, detailed, and show strong improvements based on the reviewer's feedback. Each comment is addressed professionally with specific references to the revised sections.

Response:

We are grateful for the reviewer's acknowledgment of the clarity and professionalism of our responses. The constructive feedback provided throughout the review process has been highly valuable to the team.

CS Reviewer (Dr. Khadija Safwan)

All tracked changes are seen in RED. Further improvements are found in the FDR report.

Comment 1: Fundamental Scientific and Engineering Concepts

Reviewer Comment:

Shallow description of software engineering concepts, which is expected for a non-software-centered project.

Response:

The software sections have been expanded to provide clearer architectural justification, including detailed explanations of the anomaly detection model, backend processing workflow, feature engineering strategy, and secure communication design. Greater emphasis was placed on algorithm selection rationale, real-time performance considerations, and system-level integration. These improvements were further refined and elaborated in the FDR report.

Comment 2: Project Constraints and Specifications

Reviewer Comment:

Comments were provided duringthe FDR presentation.

Response:

The specifications section has been revised to explicitly connect system constraints with design decisions. Real-time detection limits, secure communication requirements, and predictive detection objectives are now directly supported by architectural and algorithmic justifications. Additional clarification and structured linkage between constraints and implementation were incorporated in the FDR report.

Comment 3: Discipline-Specific Contributions and Team Management

Reviewer Comment:

Comments were provided duringthe FDR presentation, figure 4 is not readable

Response:

The system architecture figure has been redesigned to improve clarity, readability, and subsystem separation. Visual elements were simplified and reorganized to clearly illustrate EE–CS boundaries and backend processing flow. These revisions are reflected in the updated FDR reportt.

Comment 4: Reply to Scope Form and CDR Presentation Reviewers' Comments

Reviewer Comment:

Not all SWE concerns are resolved, but moving in the right direction

Response:

Additional refinements were made to strengthen the software engineering depth of the project, including clearer documentation of the anomaly detection approach, backend architecture, and security implementation. The FDR report further expands on these updates to address the remaining SWE concerns in a more structured and technically detailed manner.

Appendix G
Point-By-Point Reply to Final Design Presentation Reviewers' Comments

EE Reviewer (Dr. Junaid Ur Rehman)

All Electrical Engineering revisions in the written report are highlighted in **Blue using Track Changes**.

> Comment 1: Final Design
>
> Final block diagram was presented and discussed. Discussing inter-dependency of department's contribution could improve it.

Response:

During the presentation (Slide 4 – "Block Diagram with All Disciplines Contributions"

FDR Presentation Team F040), interdependencies were shown visually but not sufficiently explained verbally.

To address this:

- The written report (Section 4.1 and 4.2 EE Part) now explicitly explains:
    - How PETE leak modeling defines optical path assumptions.
    - How EE signal stability determines AI classification reliability.
    - How CS alert logic depends on EE sampling and response time.
- Figure 9 description was strengthened to clearly state cross-discipline signal and data flow.

This revision ensures that the final design is presented as an integrated multidisciplinary system rather than parallel contributions.

Comment 2: Discipline Final Design Details

Some values used in the calculation appear arbitrary (e.g., alert line calculation).

Response:

In the presentation (Slides 11–13: "Electrical Engineering – Final Design Calculations"

FDR Presentation Team F040), several values were shown without full derivation explanation.

To resolve this:

- The written report Section 4.3.1 now includes:
    - Explicit derivation of sampling rate (10 Hz) based on plume dynamics.
    - Justification of ≤ 2 s detection time using 10-sample averaging.
    - SNR reasoning for ADC resolution.
    - Explicit LEL-based threshold derivation for alert logic.
- Assumptions (temperature stability, detector linearity, quantization margin) are now clearly stated.

All numerical values are now physically justified and traceable.

Comment 3: Specifications Attainment

Some calculations seem arbitrary in demonstrating attainment.

Response:

Slides 22–26 ("Summary Table – Specifications & Constraints"

FDR Presentation Team F040, showed status (Met / Not Met Yet) but lacked full analytical justification during presentation.

To strengthen this:

- Section 4.3 – Final Design Performance Results now explicitly links:
    - Each EE specification → Supporting equation → Evidence.
- Table 4 in the written report now includes:
    - Clear section references.
    - Analytical evidence (e.g., ADC timing logs, packet loss formula, power calculation).
- LTE latency and packet loss calculations are explicitly derived rather than asserted.

This ensures specification attainment is quantitatively defendable.

Overall Comment

Ensure calculations and design decisions are well-motivated and explainable.

Response:

In response, we enhanced the EE documentation by:

- Expanding equation derivations.

- Adding explicit engineering assumptions.
- Explaining trade-offs (sampling rate vs power consumption).
- Clarifying LEL-based safety margins.
- Strengthening physical reasoning behind 1650 nm wavelength selection.

These improvements increase analytical transparency and eliminate ambiguity in design decisions.

CS Reviewer (Dr. Khadijah Safwan)

All Computer Science revisions in the written report are highlighted in **Pink using Track Changes**.

Comment 1: Final Design

document what each unit does clearly.

Response:

The final design report has been revised to clearly define the function of each CS unit within the system architecture. The Data Acquisition section now explicitly explains the EE–CS interface and absorbance transformation. The Signal Processing section details preprocessing and temporal feature extraction. The AI section clearly defines the Isolation Forest anomaly detection model and multi-level alert logic (Normal, Yellow, Red). The Backend section specifies real-time inference responsibilities, while the Security section clarifies TLS-based encrypted communication. Each unit now has a defined functional role within the predictive methane detection pipeline.

Comment 2: Discipline Final Design Details

Great design. Explore more: how are you going to exchange the AES128 key between your embedded system and the FastAPI?

Response:

The design was updated to clarify that secure communication is implemented using HTTPS with Transport Layer Security (TLS). TLS performs authenticated key exchange during the handshake phase and automatically establishes a session key. AES-based symmetric encryption is then used internally within the TLS session. Therefore, manual AES key exchange between the embedded system and FastAPI is not required, as secure key negotiation and encryption are handled by the TLS protocol itself. This approach simplifies key management while maintaining confidentiality and integrity of transmitted data.

Comment 3: Specifications Attainment

There is a lack of design around your specifications and constraints

Response:

The CS sections have been revised to explicitly connect design decisions to system specifications. The Isolation Forest model was selected due to its computational efficiency and suitability for rare-event anomaly detection, ensuring real-time inference within the ≤2 second constraint. The predictive multi-level alert logic supports early warning capability as required by system safety objectives. HTTPS-based encrypted transmission satisfies data security requirements. Additionally, backend deployment allows model updates without firmware changes, improving maintainability and scalability. Each specification is now supported by a corresponding architectural or algorithmic justification.

Overall Comment

Avoid reading unless presenting technical terms or numbers. There are a couple of misspellings.

Response:

Thank you for the feedback. We will avoid reading during presentations and focus on clear conceptual explanation. The report has also been reviewed to correct spelling and formatting issues.

Coach (Dr. Umer Zahid)

Comment 1: Elevator Pitch

Elevator pitch needs significant improvement in terms of delivery and speech. Current pitch is too long where student was just reading it with her tablet. The purpose of pitch is to present the theme/idea of the project in a non-technical and persuasive manner.

Response:

The elevator pitch has been revised to be shorter, clearer, and more persuasive. Technical details were minimized to focus on the core idea, problem motivation, and impact. The delivery approach has been improved to avoid reading directly from notes and instead emphasize confident, conceptual presentation of the project theme.

Comment 2: Gantt Chart and Project Timeline Adherence

Gantt chart is well constructed, however, difficult to read because of poor resolution. Also, the timeline is fragmented and many tasks not mapped on the timeline.

Response:

The Gantt chart has been redesigned with improved resolution and formatting to enhance readability. Tasks have been consolidated and clearly mapped onto the timeline to reflect parallel execution across disciplines and better adherence to project milestones.

Comment 3: Deliverables

Deliverables should be compact and precise about the final product(s). Many of the deliverables seems like tasks. Putting lengthy descriptions dilute the main focus of the work.

Response:

The deliverables section has been revised to focus on clear, measurable final products rather than intermediate tasks. Descriptions were condensed to highlight the tangible outputs of the project, including the sensing prototype, validated dataset, anomaly detection module, and real-time monitoring interface.

Comment 4: Visual Support and Delivery

Delivery and visual support is good.

Response:

Thank you for the positive feedback. Efforts will continue to maintain clear visual communication and structured presentation flow.

Comment 5: Overall Comments

Based on the reviewer comments, overall, the project is moving in the right direction. Students are recommended to get continuous feedback from respective evaluators.

Response:

We appreciate the feedback and will continue seeking regular guidance from evaluators to ensure steady improvement in both technical development and presentation quality.

PETE Reviewer (Dr. Duaa Abdelwahab)

All Petroleum Engineering revisions in the written report are highlighted in **Red using Track Changes**.

Comment 1: Final Design

"Block diagram is mostly clear and well-structured."

Response:

Thank you for the positive feedback regarding the clarity and structure of the block diagram.

Comment 2: Discipline Final Design Details

"PE contributions are solid in defining leak severity levels. You should expand your role by adding quantative leakage PE analysis, the pressure dependent leakage, dispersion behavior... etc. In addition, applying the developed system to a field case would significantly strengthen the PE contribution by demonstrating methane leak detection at different surface facility components such as wellheads, chokes, pipelines, and related surface equipment."

Response:

Thank you for the valuable recommendation. In response, the Petroleum Engineering contribution has been expanded significantly in Sections 4.2 and 4.3.1. Quantitative analysis was incorporated through LEL normalization, methane concentration-to-%LEL conversion, and a structured severity classification framework grounded in flammability physics. Additionally, a dispersion-based interpretation model was added to explain concentration decay relative to leak proximity. The discussion was further strengthened by incorporating realistic field-relevant leakage scenarios at pipeline joints, flanges, and surface facility components. These additions expand the PE role beyond qualitative interpretation into quantitative safety-based analysis.

Comment 3: Specifications Attainment

"You should add evidence for specification and constraints."

Response:

Thank you for highlighting the need for clearer traceability. Table 6 has been revised to explicitly include evidence locations for all constraints, specifications, and integrated specifications. Each item now references the exact report section where it is justified, including Sections 4.2 (Detailed Design Description), 4.3.1 (Final Design Analysis), and 4.3.2 (Final Design Results). OSHA compliance, LEL threshold selection, AI classification accuracy, sampling rate, and system latency are now directly linked to documented analysis and results.

Overall Comment

"Overall, the project is very good; however, the role of Petroleum Engineering remains limited and should be expanded by incorporating the above-mentioned quantitative analyses, field-scale application, and surface facility leakage scenarios."

Response:

Thank you for the overall positive evaluation of the project. The Petroleum Engineering role has been integral from the conceptual phase, particularly in defining the safety context, risk-based detection philosophy, and engineering meaning of methane concentration measurements. PETE established the LEL-referenced detection strategy, structured the leak severity framework, ensured compliance with flammability safety standards, and aligned the system with realistic oil and gas surface infrastructure scenarios. Rather than focusing on hardware development, the PETE contribution ensured that the system produces safety-interpretable outputs grounded in petroleum engineering principles. Therefore, the discipline's role is analytical, safety-oriented, and essential to the overall functionality of the integrated prototype.